

Private Cloud Mastery: Building Resilient Infrastructure with Proxmox VE





Introduction to Cloud Computing

Setup Proxmox Virtualization

Identity and Access Management

Virtualization Networking and Storage Models

Introduction to Cloud Computing



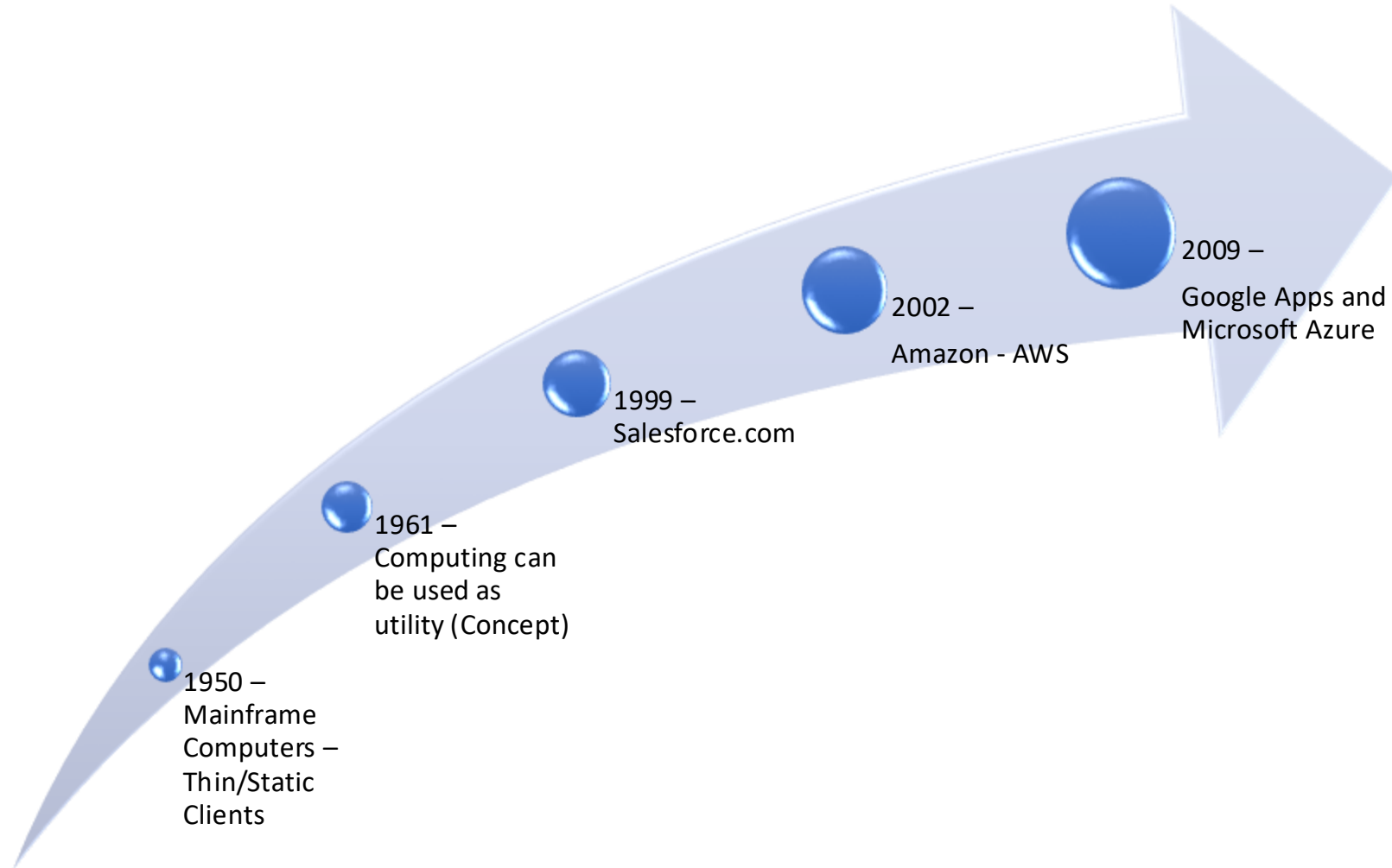
Introduction to Cloud Computing

Cloud computing is not a new technology, but an IT delivery model using existing technology. Its main purpose is to transform computing resources into a utility just like electricity or water, which can be easily accessible and charged according to its use.

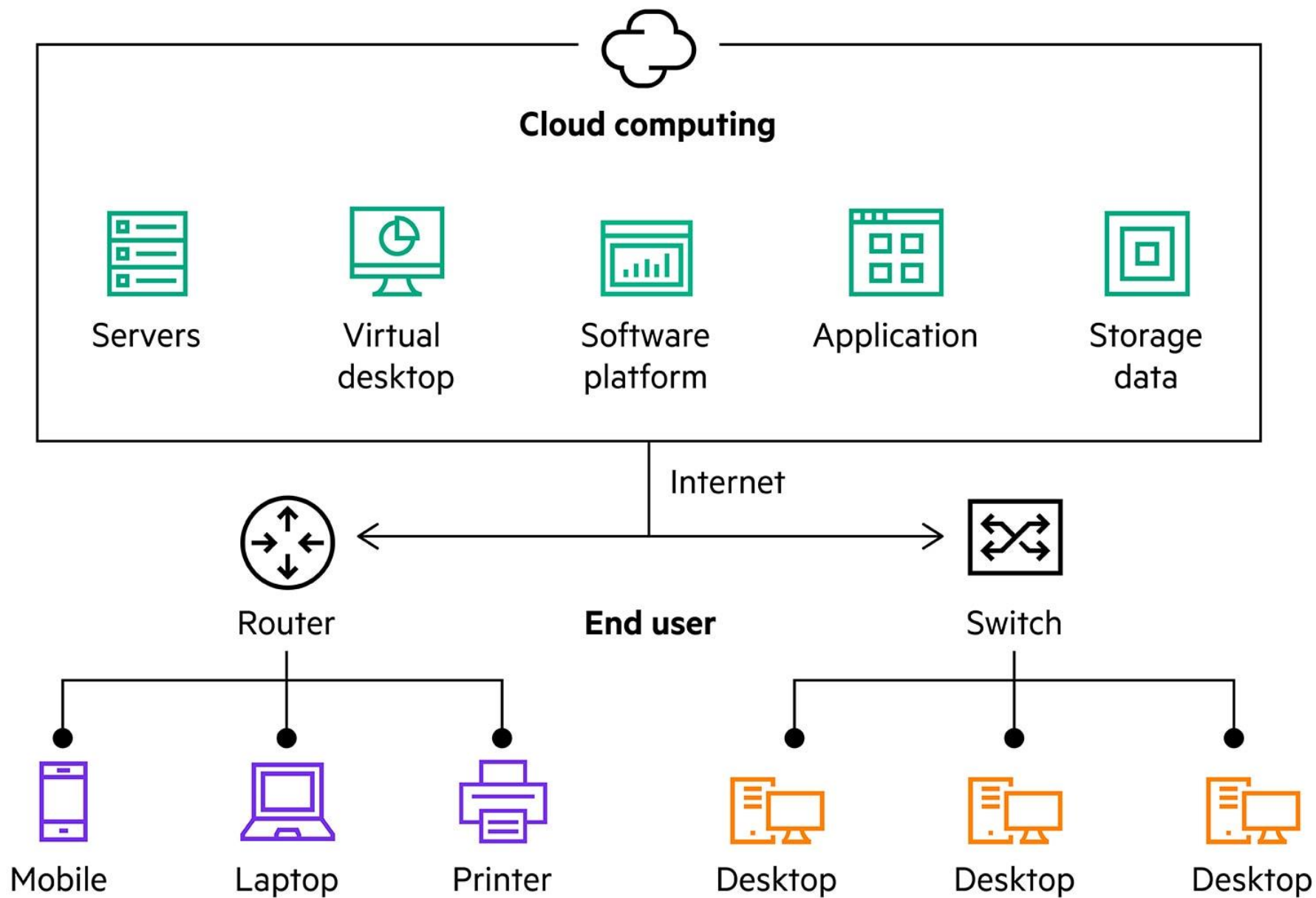
Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Cloud computing implementations provide an extra layer of abstraction (another one!) that hides the complexity of IT implementations and support from its consumers.

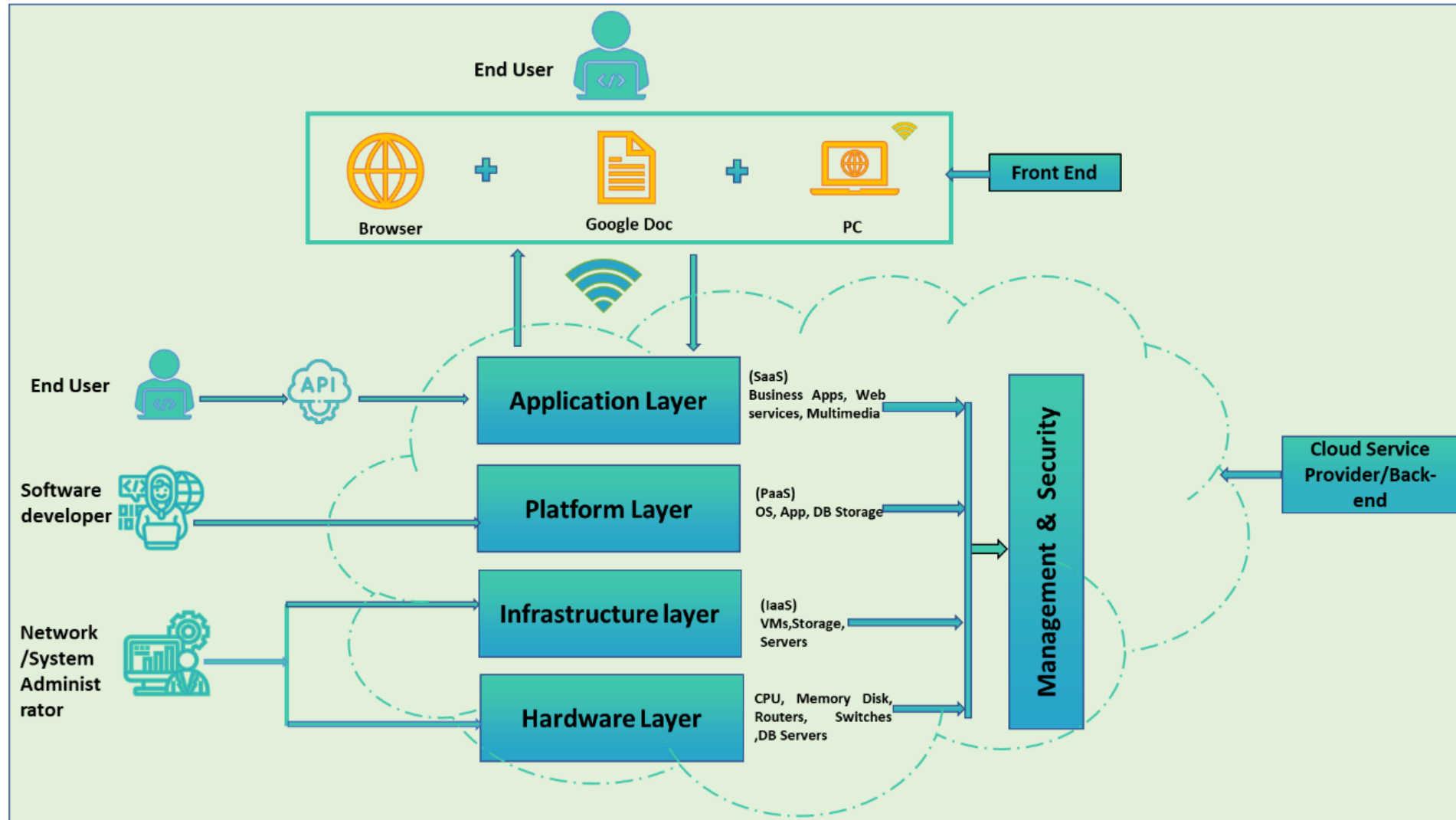
Journey of Cloud Computing



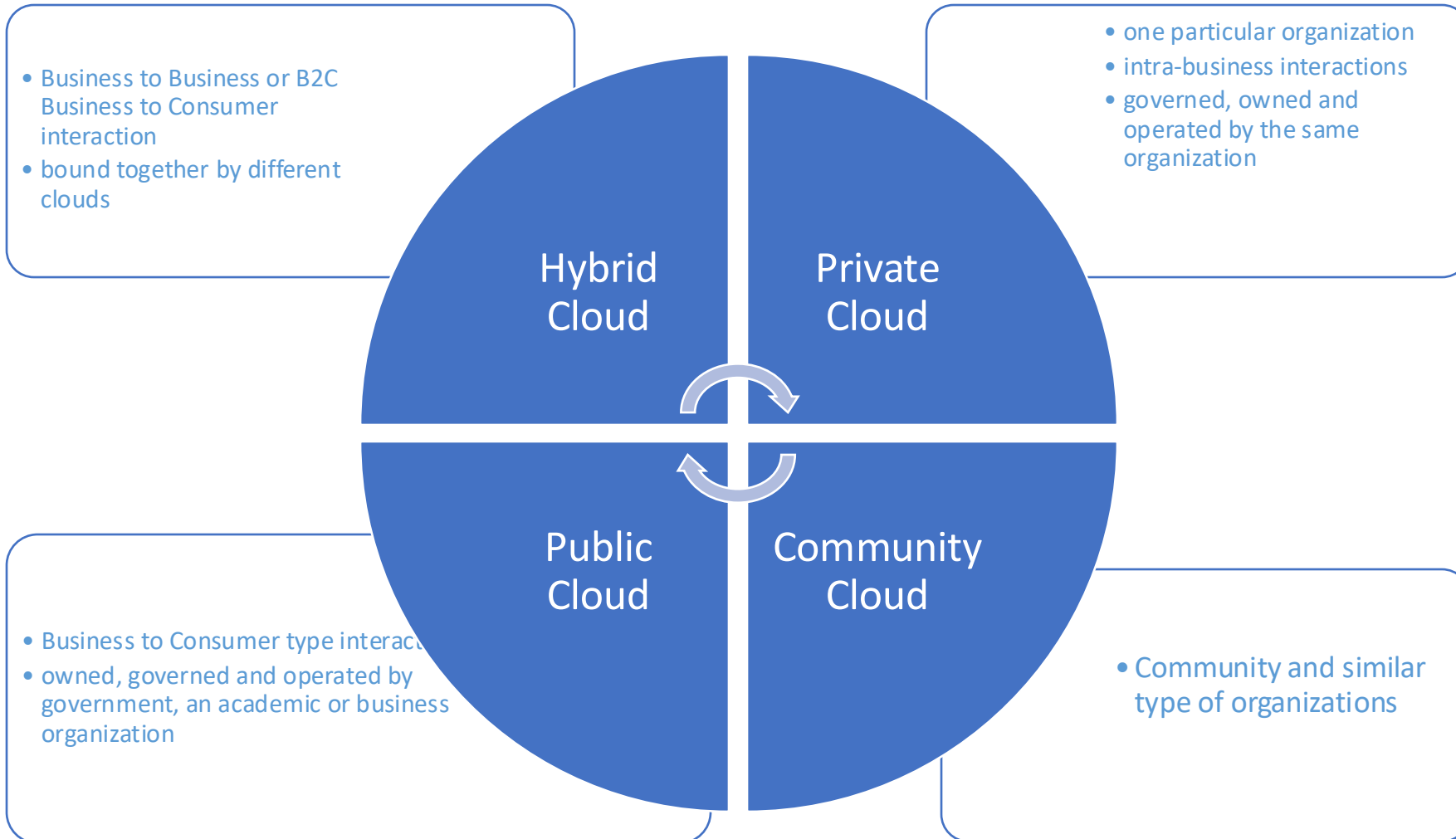
Cloud computing architecture



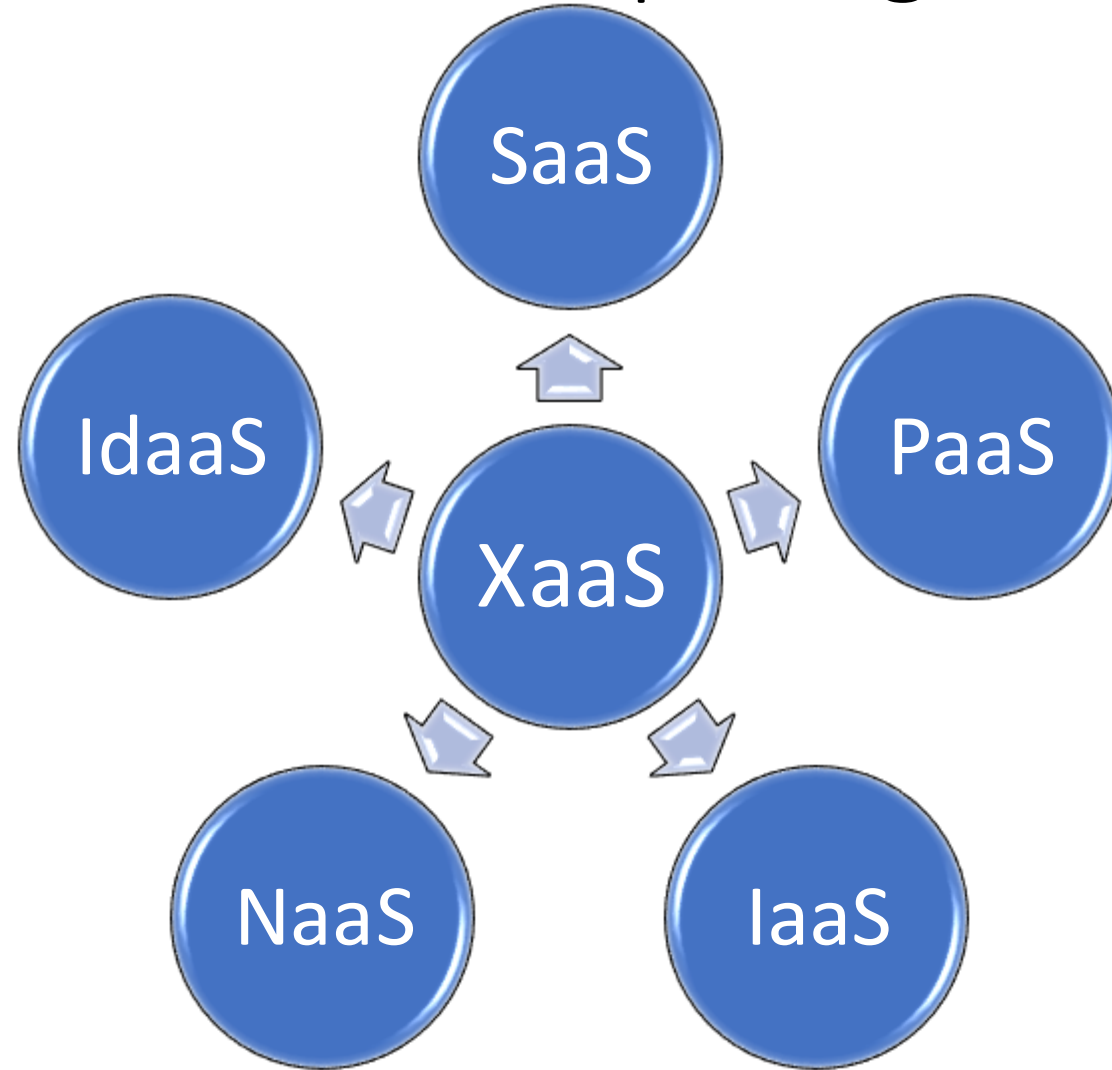
Cloud Computing Architecture



Types of Cloud Computing

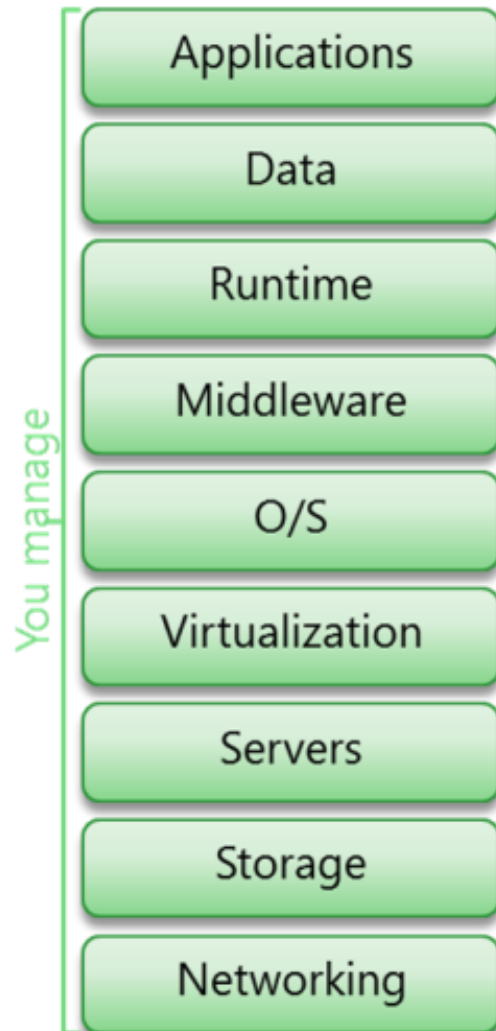


Services of Cloud Computing

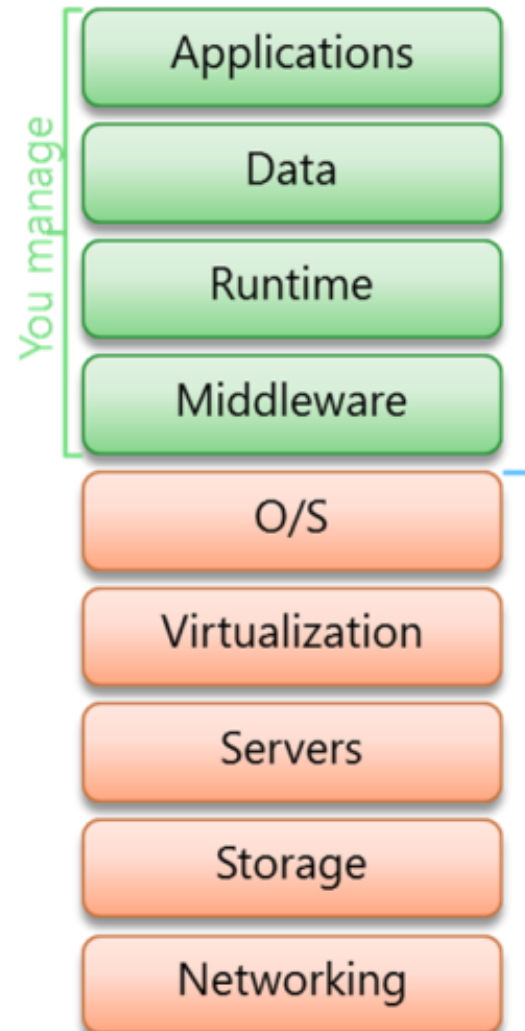




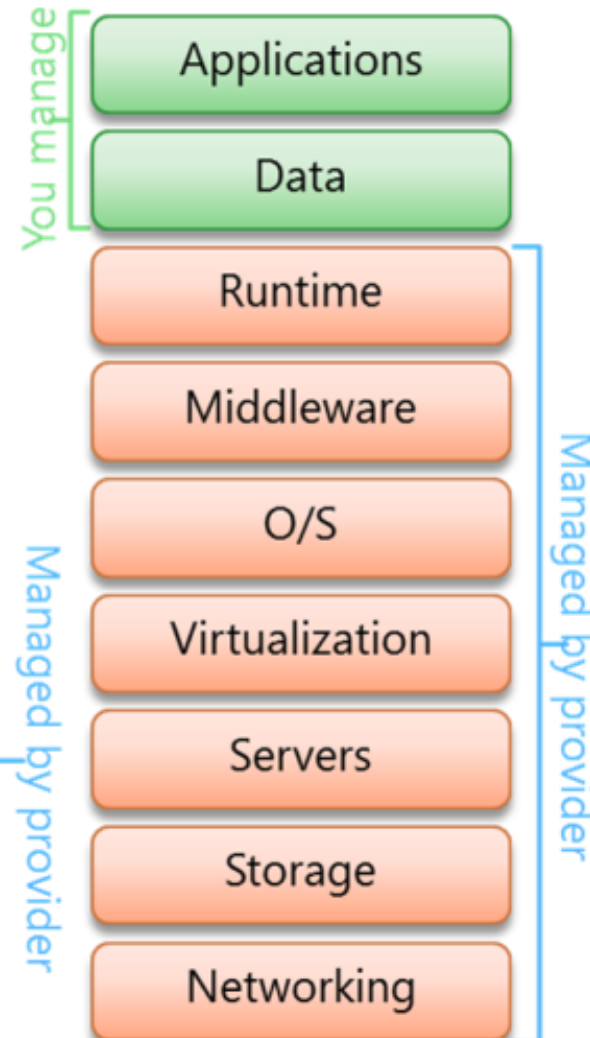
On-premise Environment



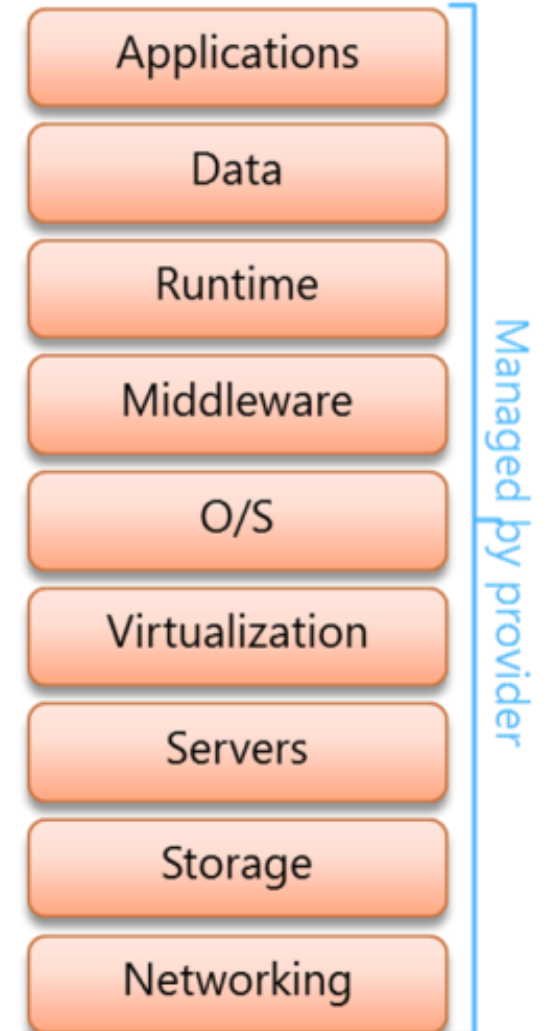
Infrastructure (as a Service)



Platform (as a Service)



Software (as a Service)



Introduction to Virtualization

Virtualization: Transparent emulation of IT resources producing benefits to consumers which is unavailable in physical form

Emulation: Main Memory, Mainframe, Default Gateway IP Address etc.

- ✓ Simulation: Act of Pretending
- ✓ Dissimulation: Act of Concealing

Transparency: CPU, Mainframe Users, TCP/IP Host etc.

Benefits:

- ✓ Memory Expansion
- ✓ Resource Optimization
- ✓ High Availability

Types of Virtualization

Pooling

(Many -> One)

- Homogeneous – Comprises of similar elements
- Heterogeneous – Comprises of different physical elements.

Partitioning

(One -> Many)

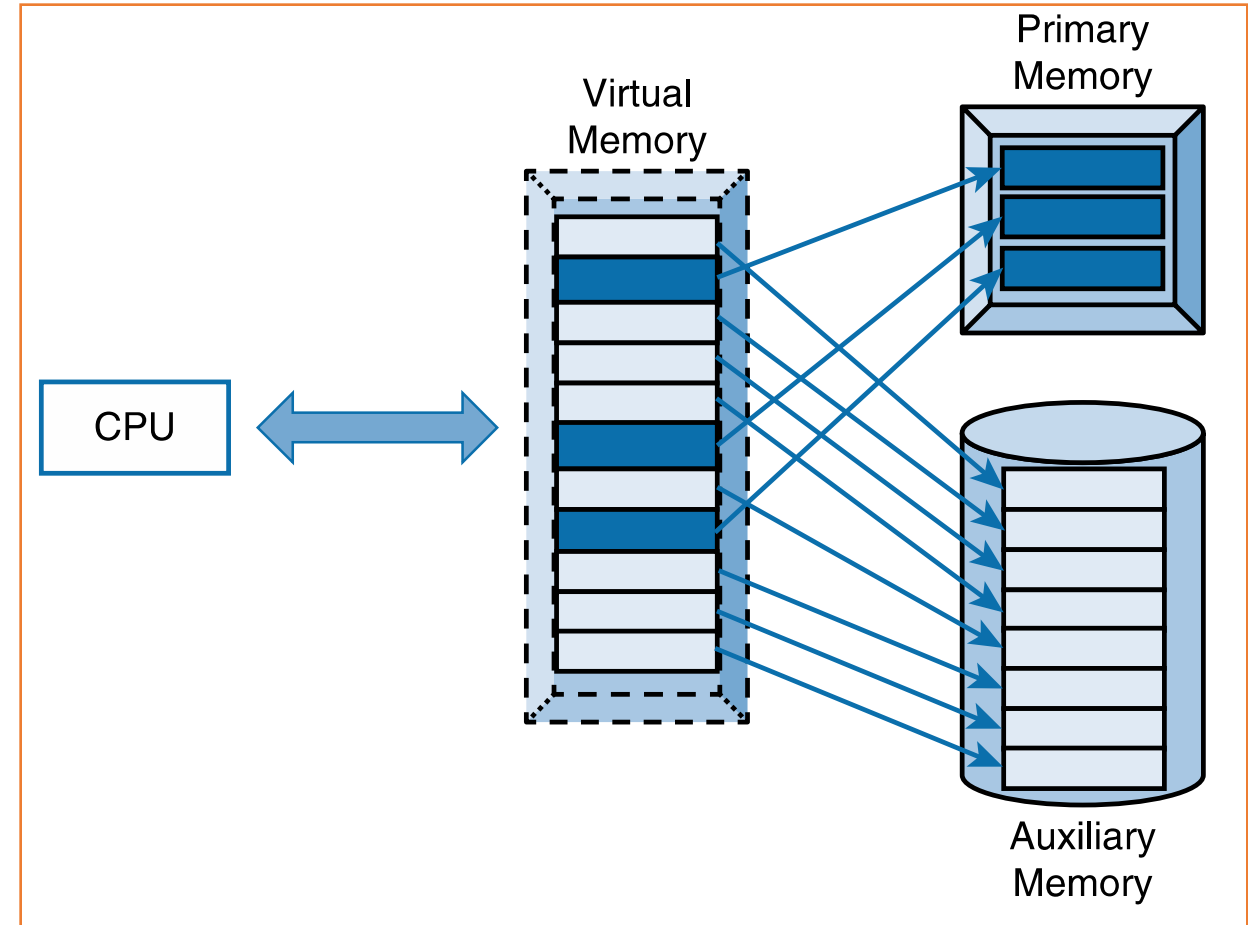
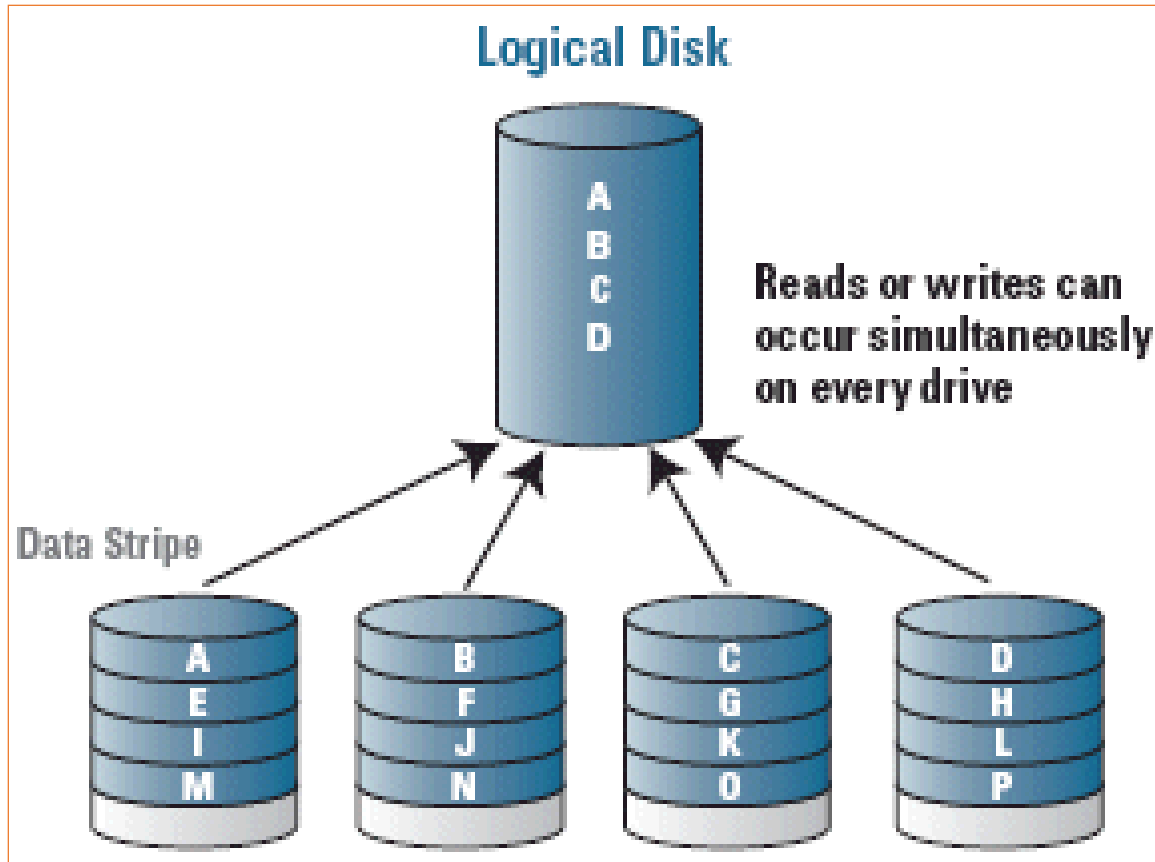
- Resource Allocation – Assets from physical resources are reserved
- No Resource Allocation – No resources are reserved

Abstraction

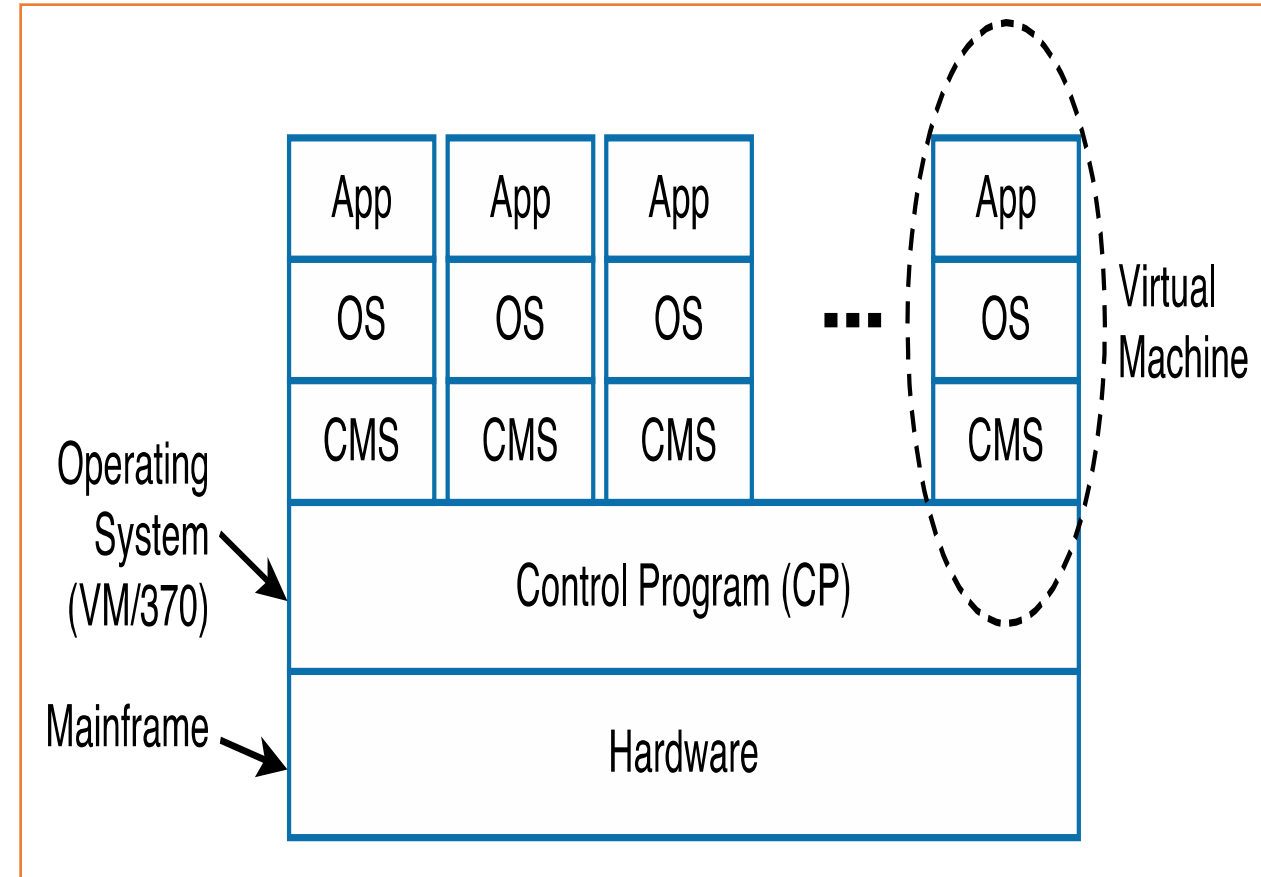
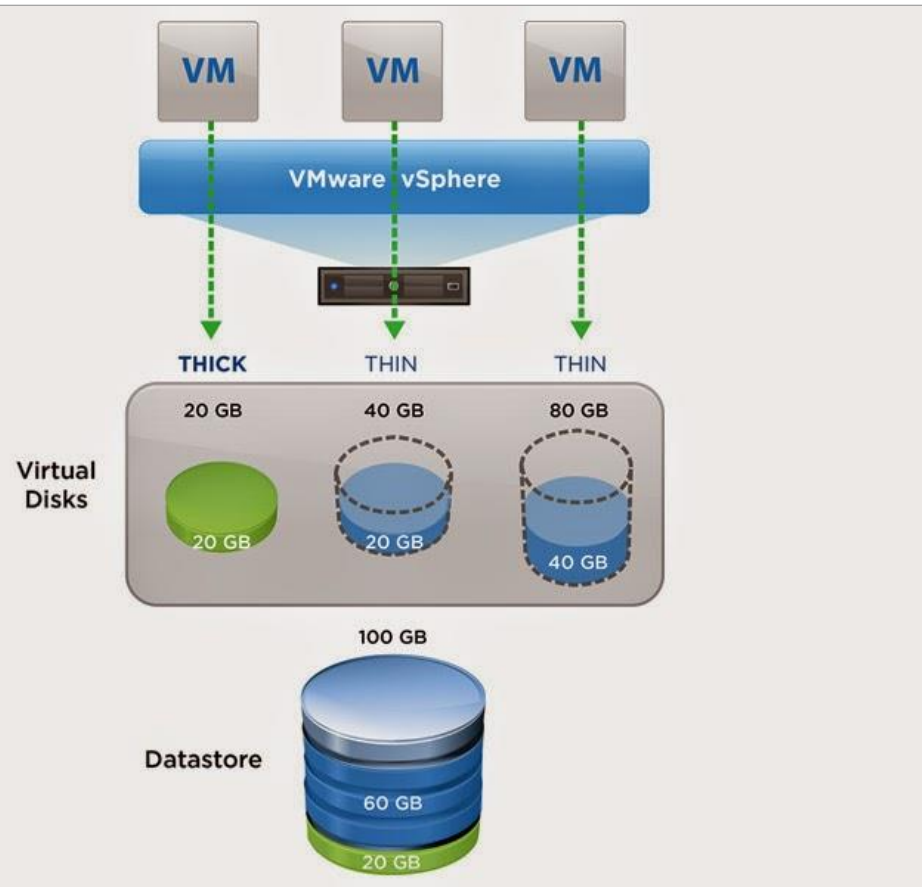
(One -> One)

- Address Remapping – Address or identities are changed
- Structural – Does not share the format of physical element

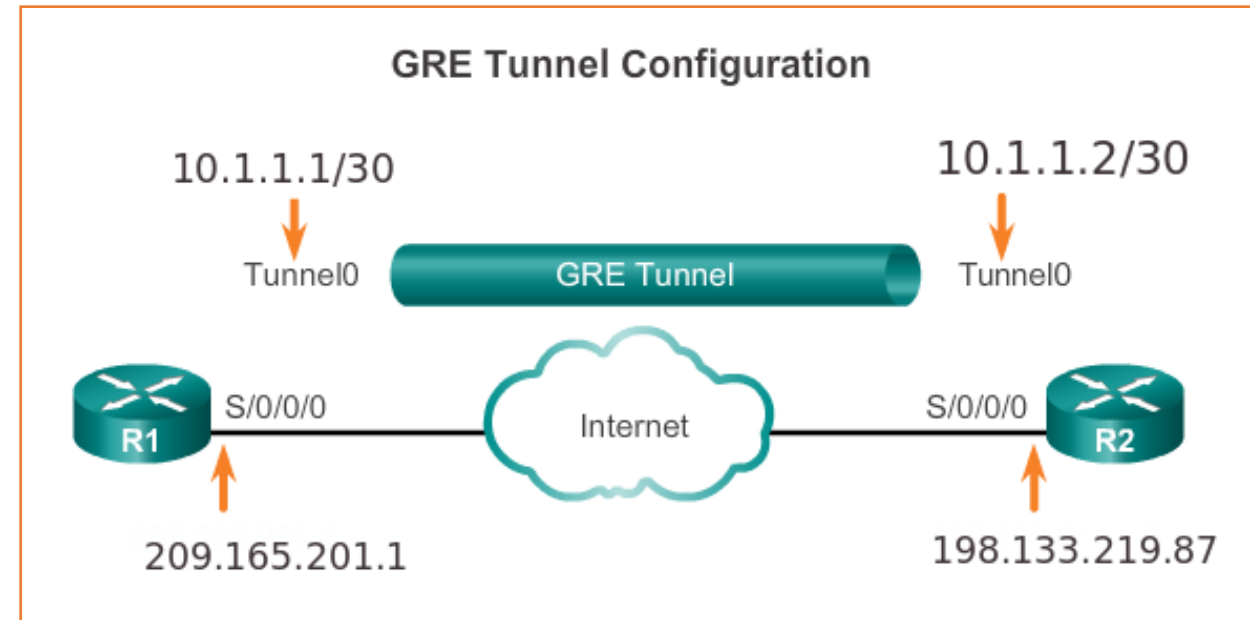
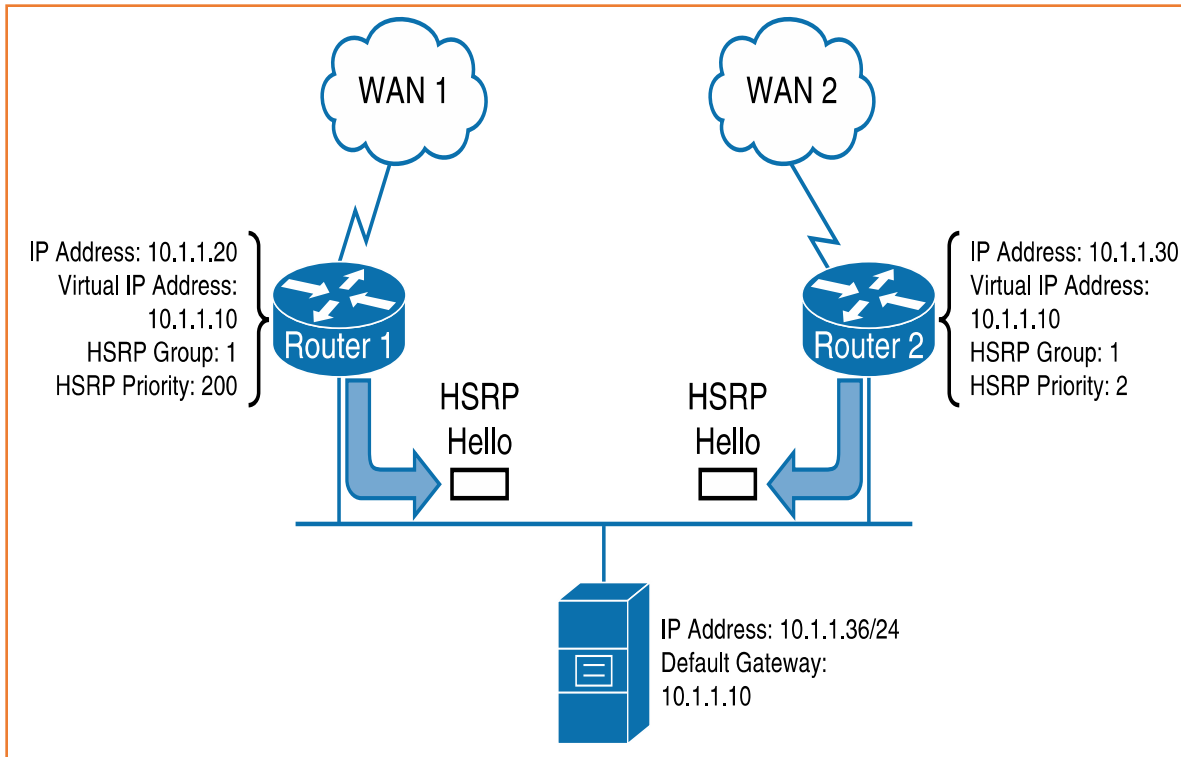
Types of Virtualization (Pooling)



Types of Virtualization (Partitioning)



Types of Virtualization (Abstraction)



Area of Virtualization

Compute / Server

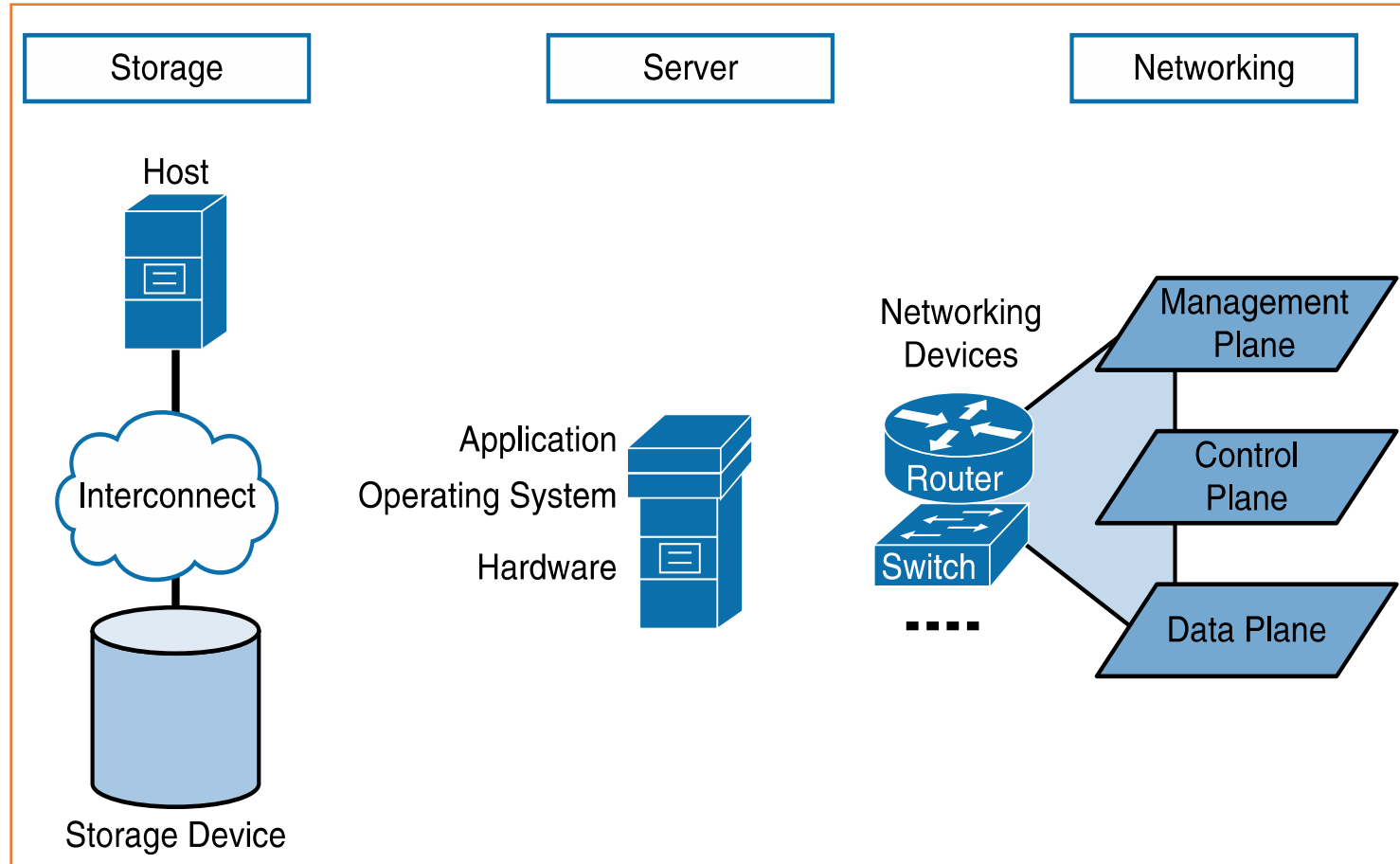
- Hardware
- Operating System
- Application

Network

- Data Plane
- Control Plane
- Management Plane

Storage

- Storage Device
- Host
- Interconnect



Server Virtualization Functionality

Hypervisors:

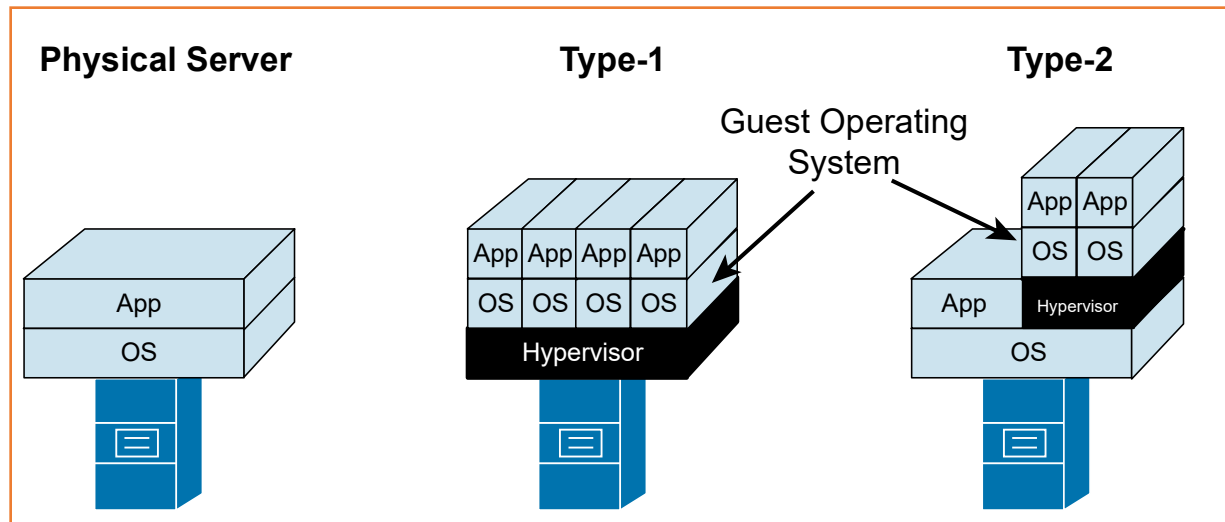
- ✓ A Software Component
- ✓ Create emulated hardware
 - CPU
 - Memory
 - Storage
 - Peripherals
- ✓ Allow to create Virtual Machine

Type - 1

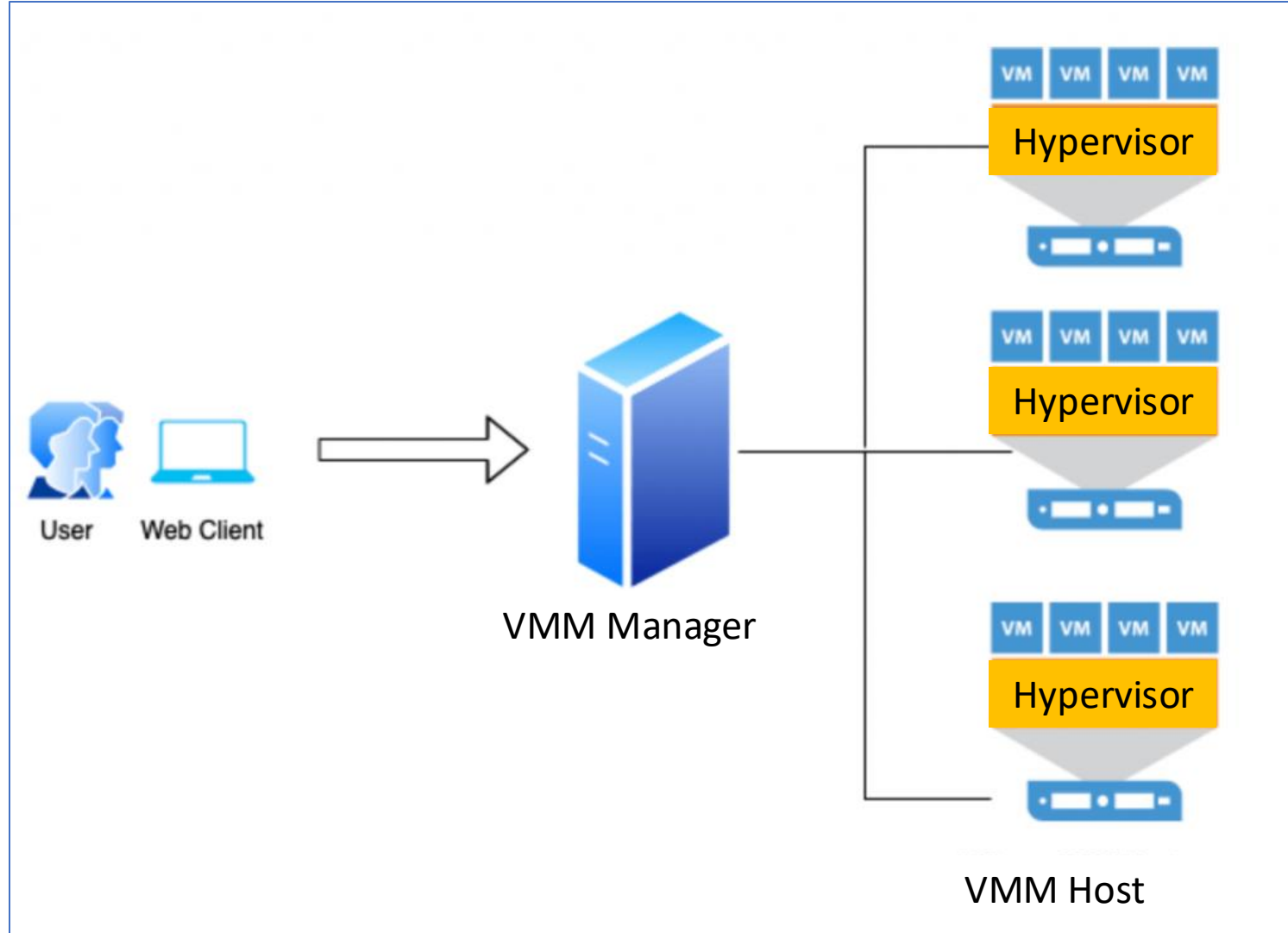
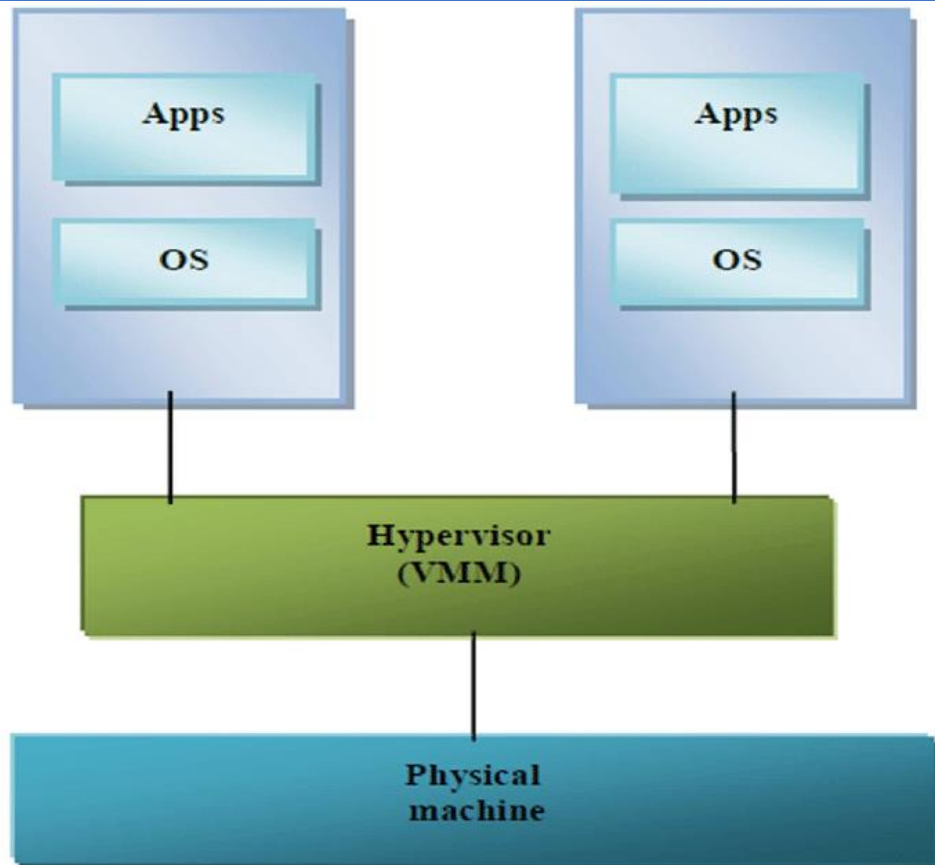
- VMware ESXi
- Microsoft Hyper-V
- Red Hat Enterprise Virtualization (RHEV)
- Citrix XenServer
- Oracle VM

Type - 2

- Linux KVM
- VMware Workstation
- VMware Player
- VMware Fusion
- Microsoft Windows Virtual PC
- Oracle VM Virtual Box
- Parallels Desktop for Mac



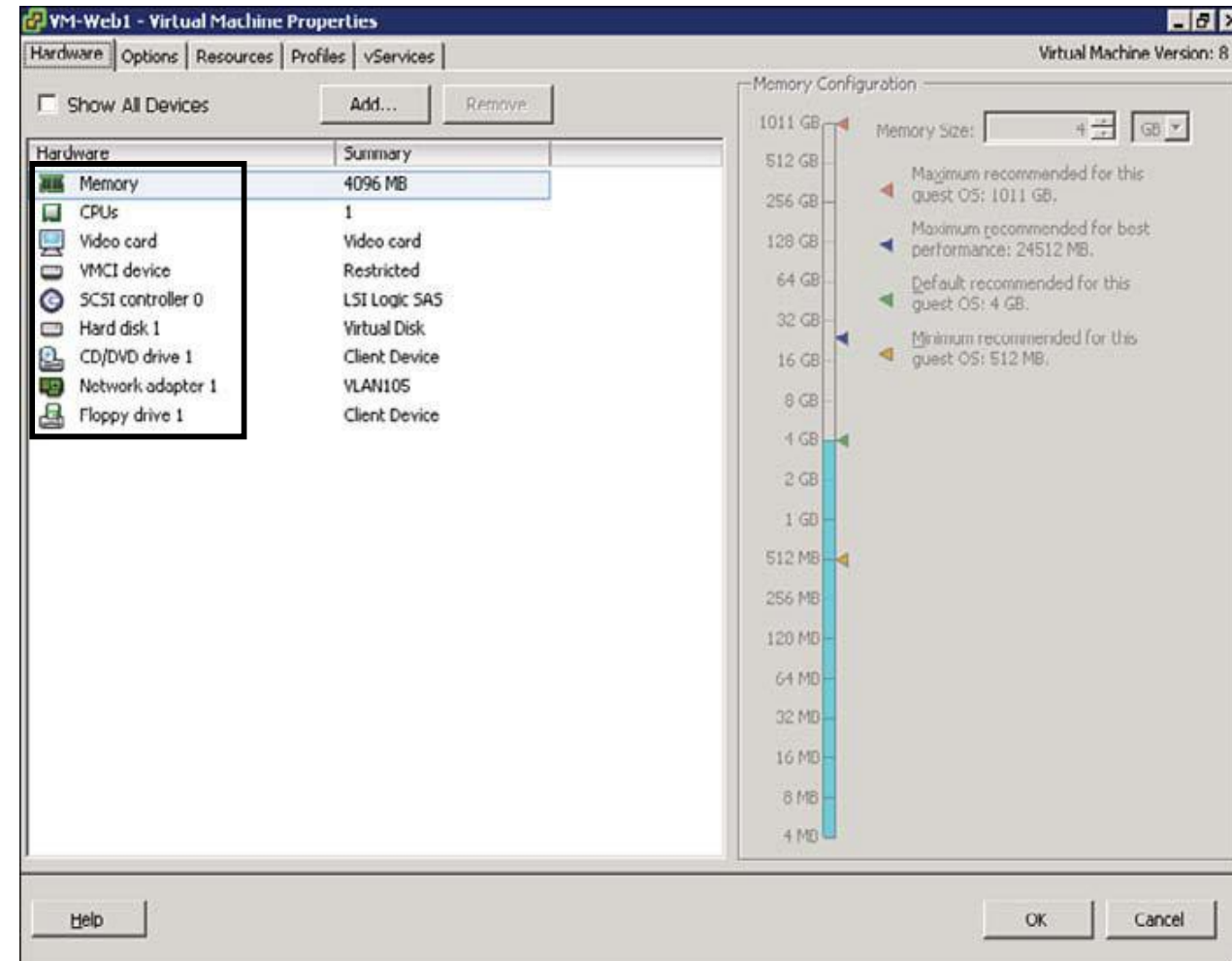
Server Virtualization Management



Server Virtualization Technology

Virtual Machine Deploys Virtual Hardware Devices

- ✓ Virtual Central Processing Unit (vCPU)
- ✓ Virtual Random Access memory (vRAM)
- ✓ Virtual Hard Drive
- ✓ Virtual Storage Controller
- ✓ Virtual Network Interface Controller (vNIC)
- ✓ Virtual Video Accelerator Card
- ✓ Virtual Peripherals – CD, DVD, Floppy Disk Drive



Server Virtualization Technology

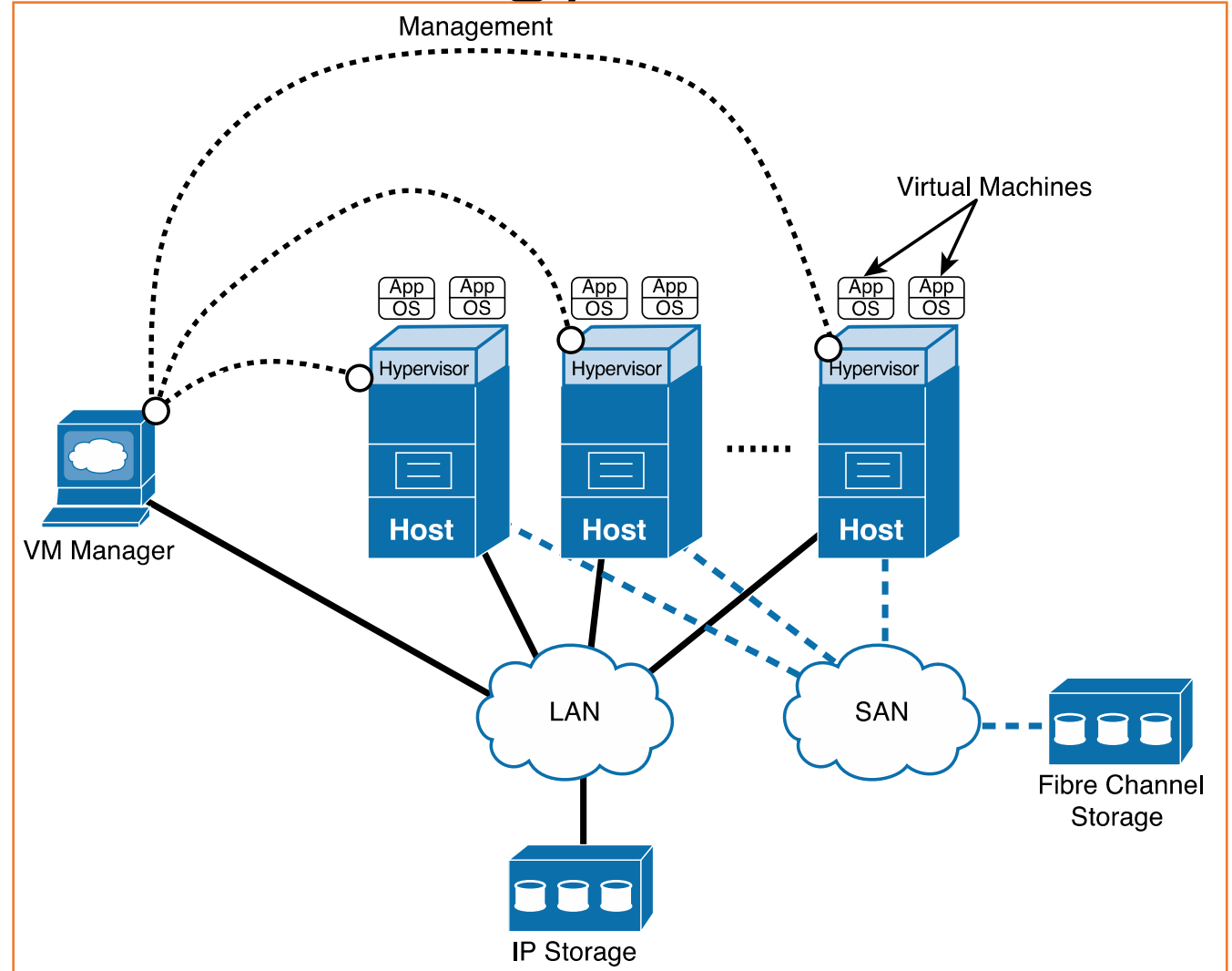
Virtual Machine is composed of a set of files which dictate how Hypervisor controls physical resources and shares them with guest OS.

File Type	File Extension	Functions
Virtual Disk	.vmdk	✓ Contains all the data ✓ VM uses as internal storage device
Swap Memory	.vswp	✓ Used as a replacement of virtual memory
Log	.log	✓ Stores all the information ✓ Used for troubleshooting purposes
Configuration	.vmx	✓ Contains all information of hardware settings ✓ vRAM Size, NIC information etc.
Nonvolatile RAM	.nvram	✓ Contains information for VM initialization ✓ Boot order, CPU usage etc.

Server Virtualization Technology

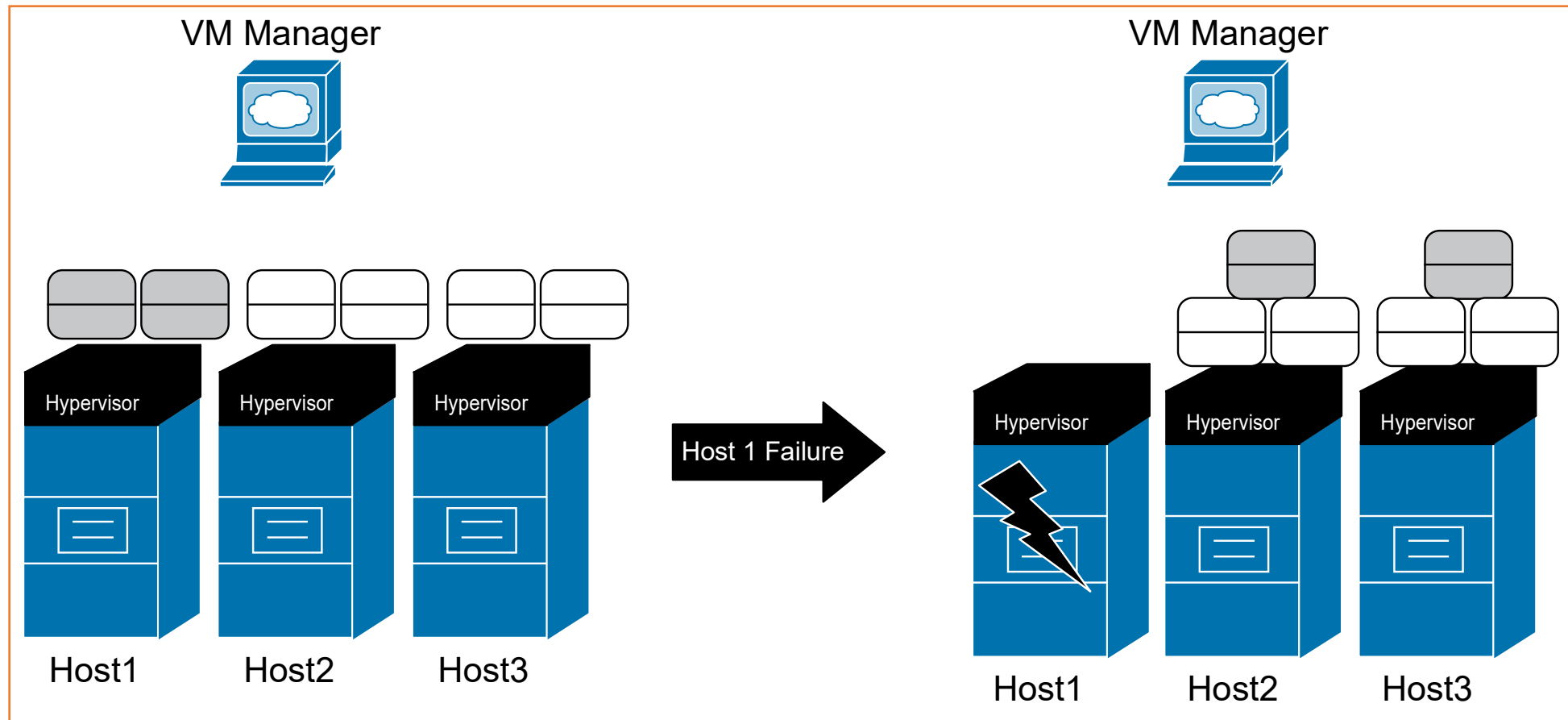
Virtual Machine Manager:

- ✓ Software Solution
- ✓ Create and Manage Virtual Machine



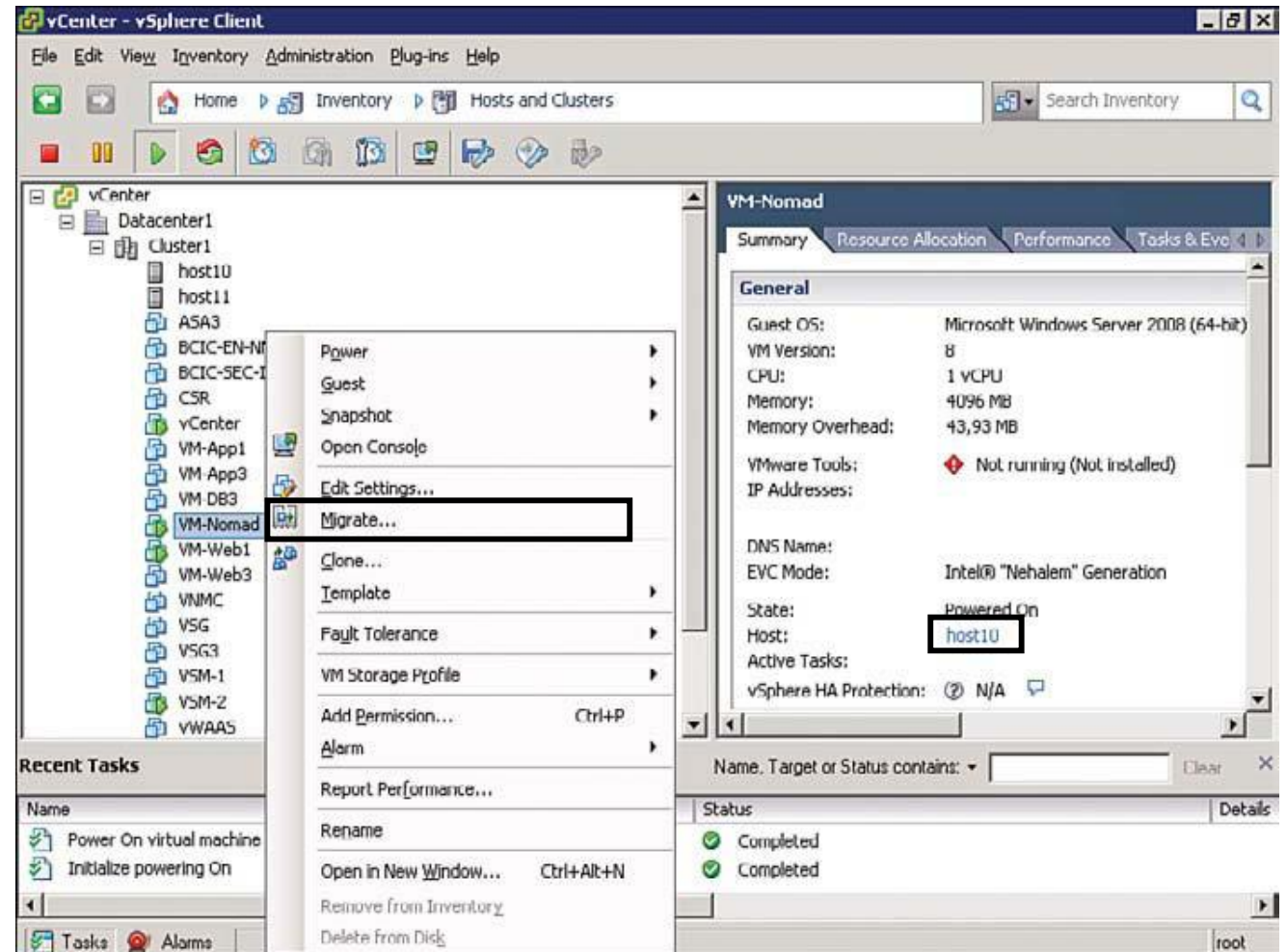
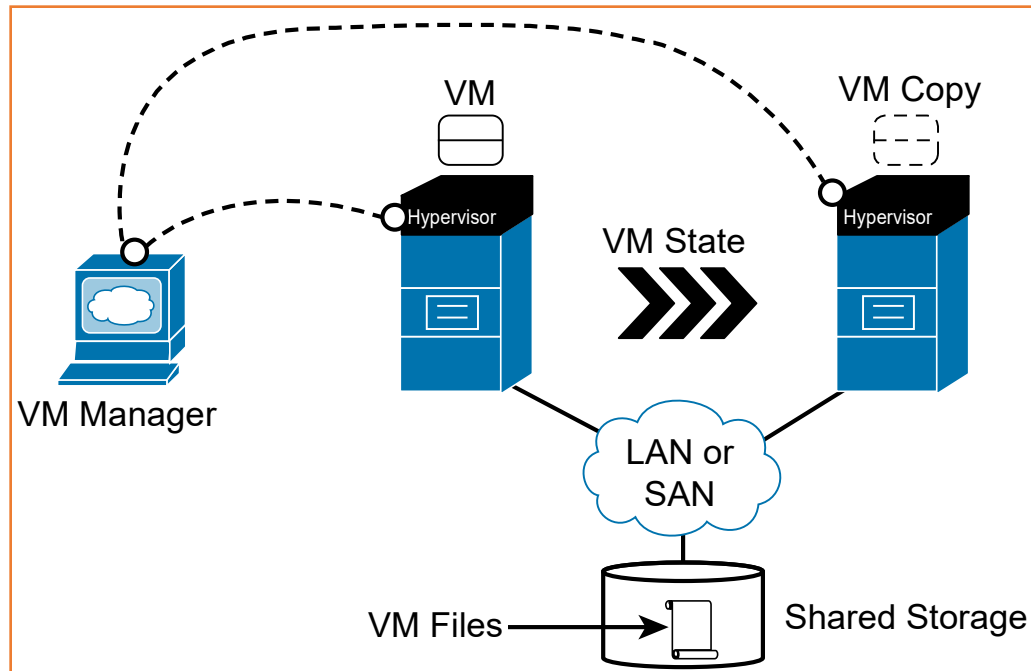
Server Virtualization Features

Virtualization Machine High Availability:



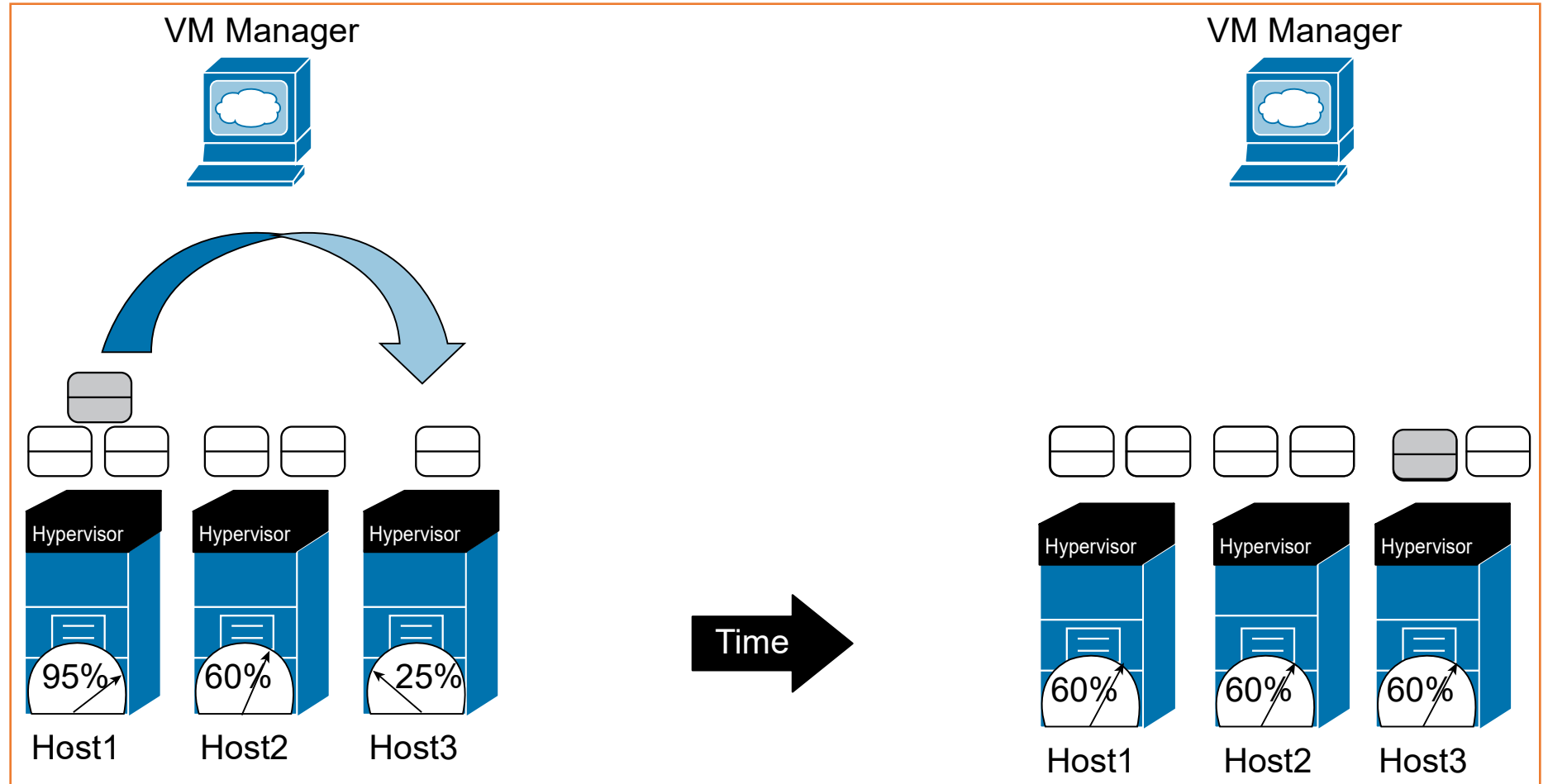
Server Virtualization Features

Virtualization Machine Live Migration:



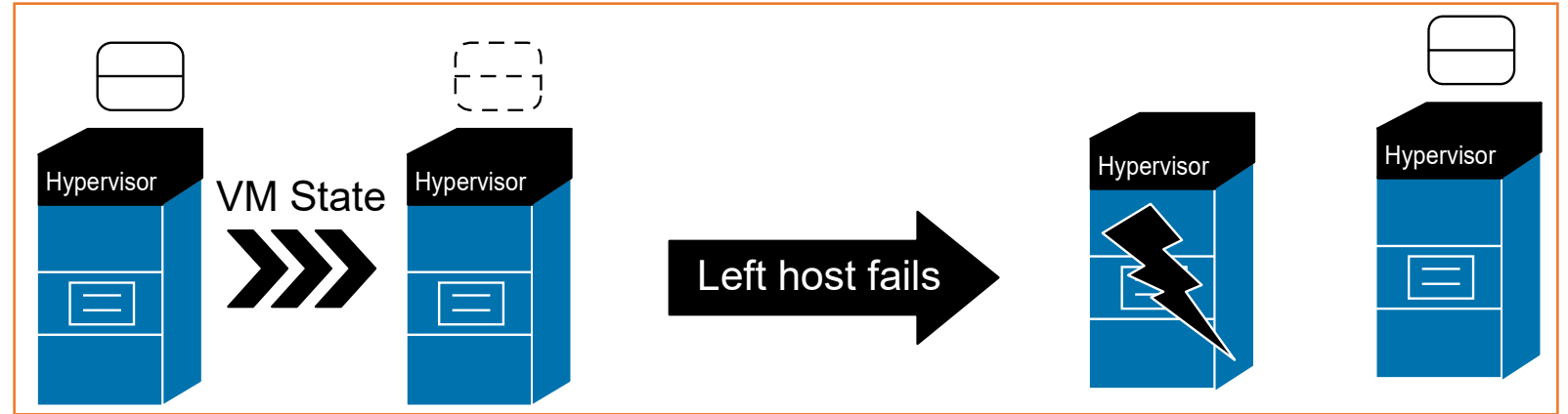
Server Virtualization Features

Resource Load Balancing :

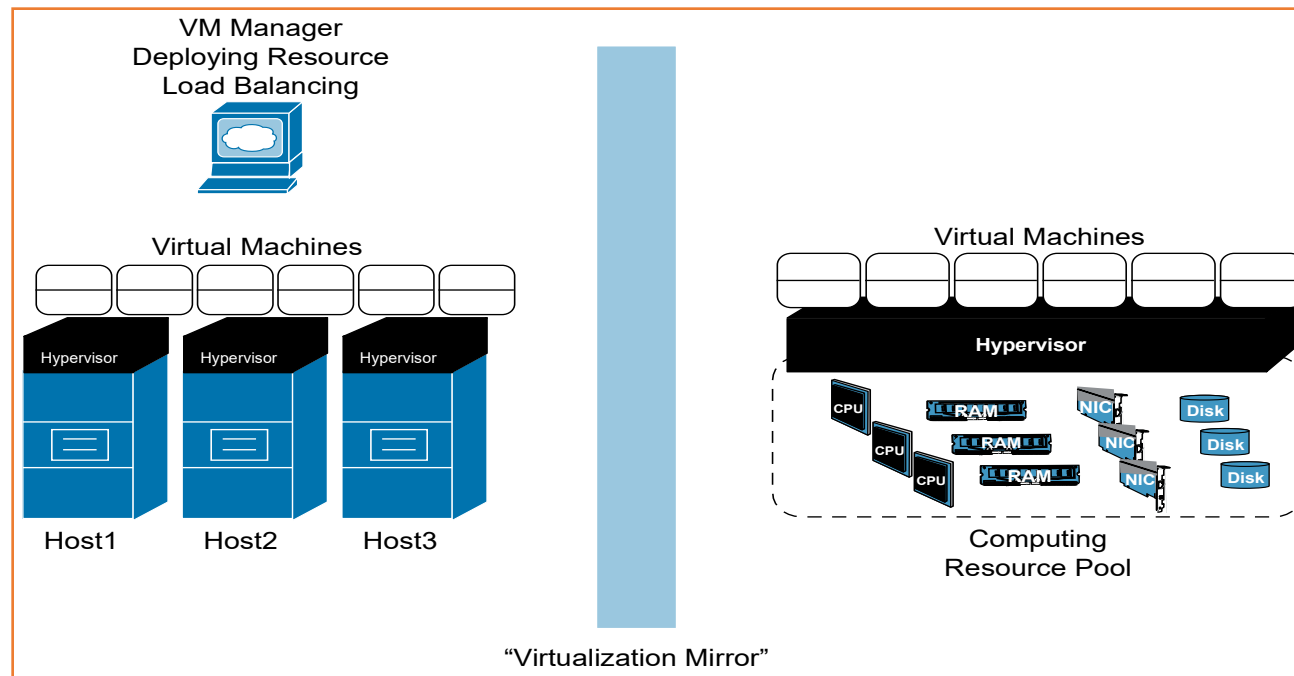


Server Virtualization Features

Virtual Machine Fault Tolerance :



Resource Pooling :



- Power Management
- Maintenance Mode
- Snapshot
- Cloning
- Templates

Desktop Virtualization Infrastructure (VDI)

- Virtual Desktop Infrastructure (VDI) is a virtualization technology that hosts a Desktop Operating System on a Centralized Server in Data Center
- A Variation of Client-Server Computing Model
- Sometimes referred to as Server-Based Computing

Types

- Persistent
- Non-persistent

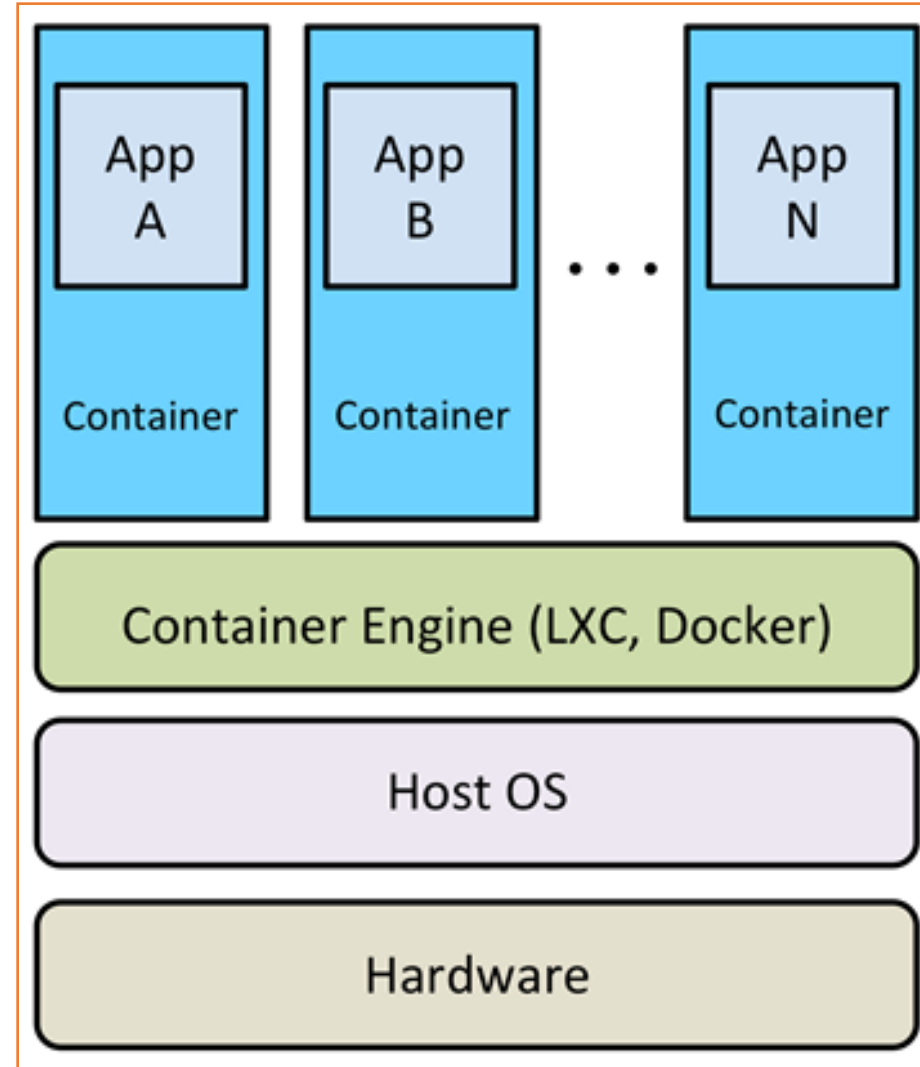
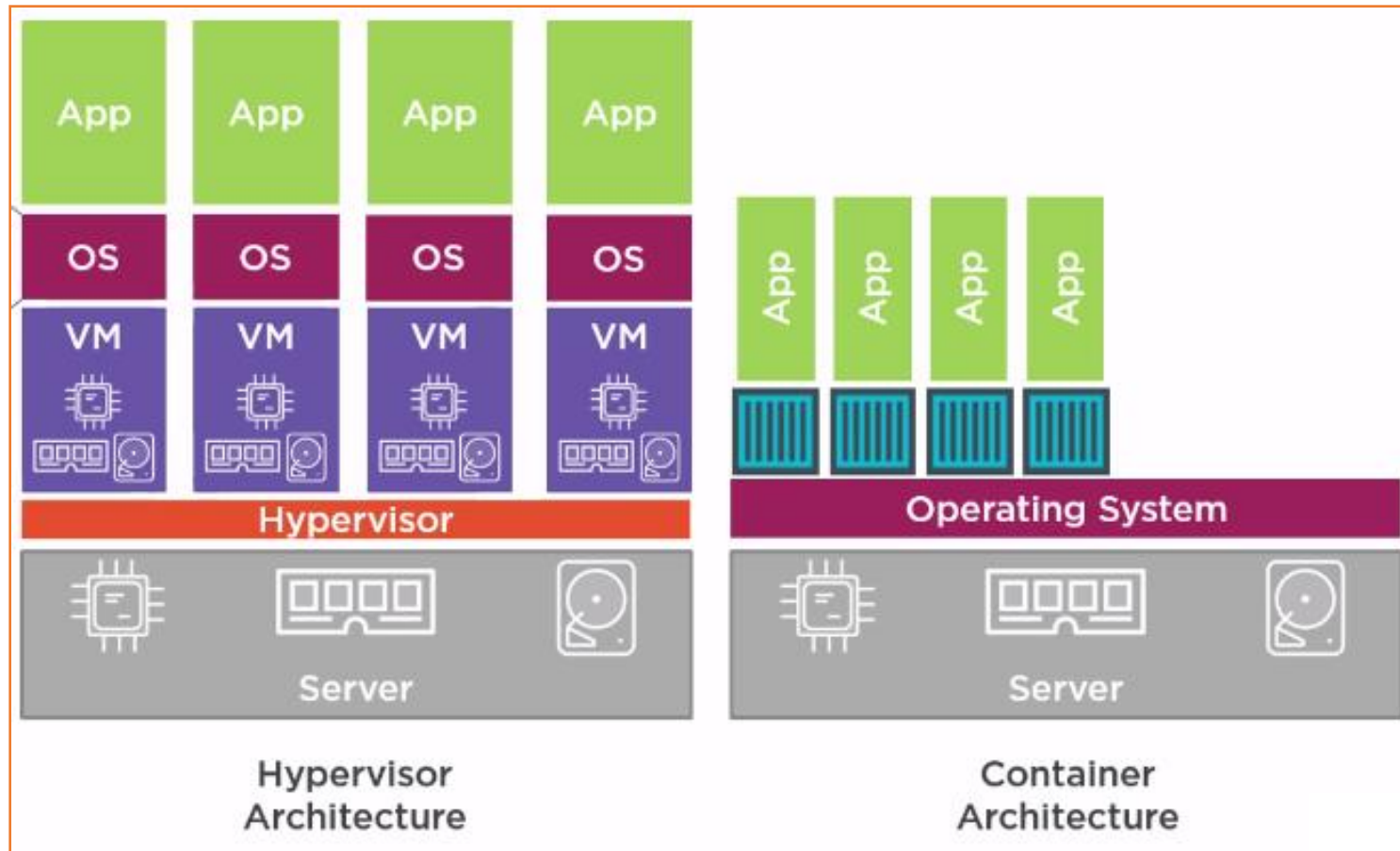
Usage

- Remote Work
- Flexibility / BYOD
- Security
- Task / Shift Duty

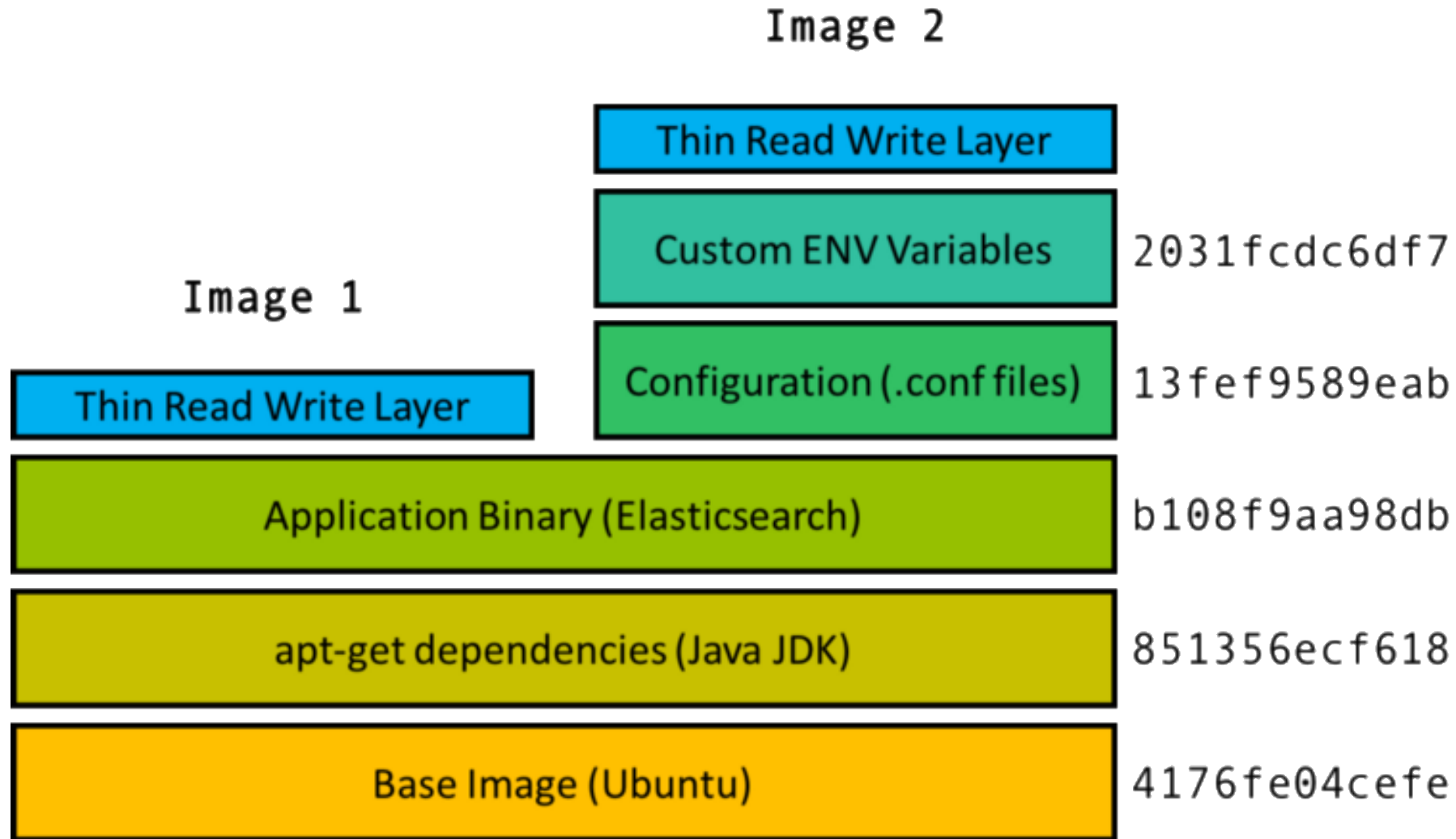
Benefits

- Mobility
- Cost Savings
- Security
- Centralized Management

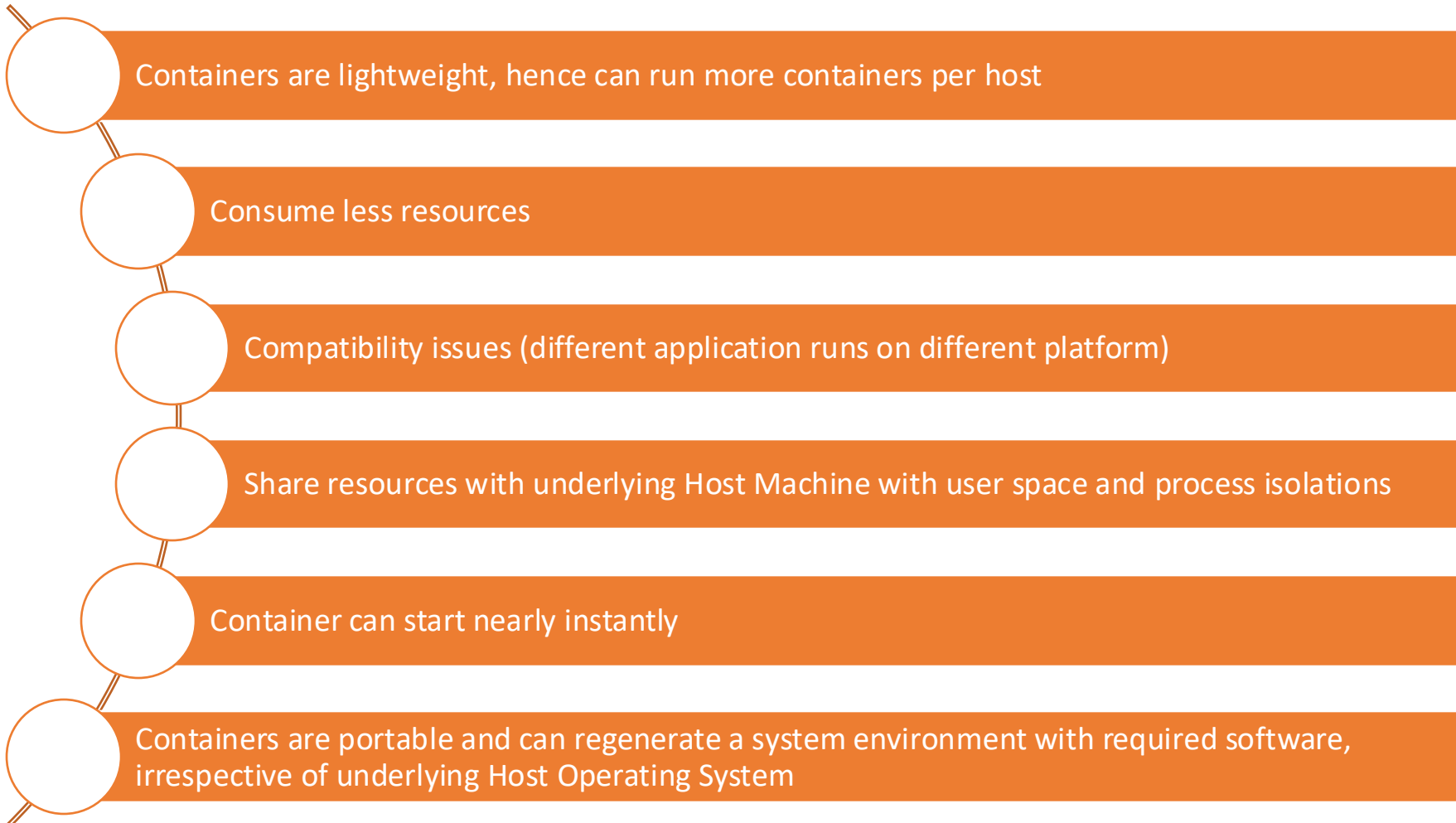
Hypervisor and Container Architecture



Architecture of Container

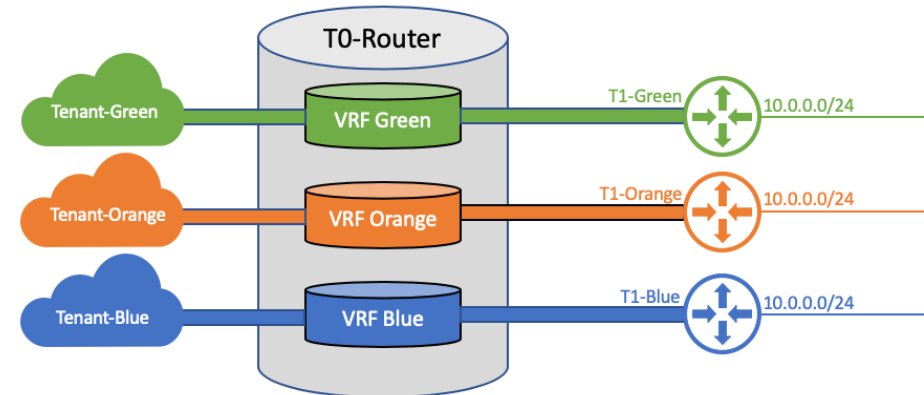
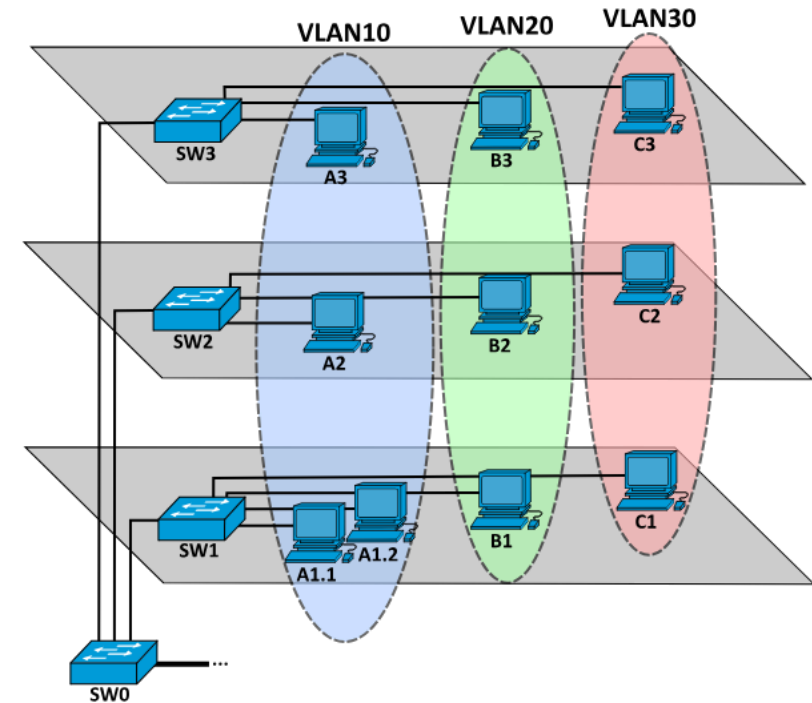


Benefits of Container Compared to VM

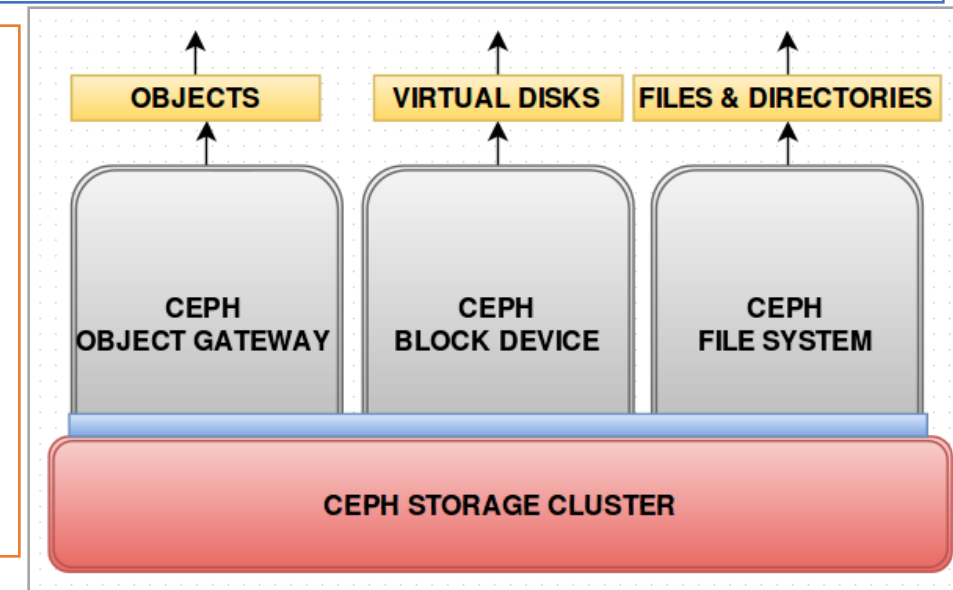
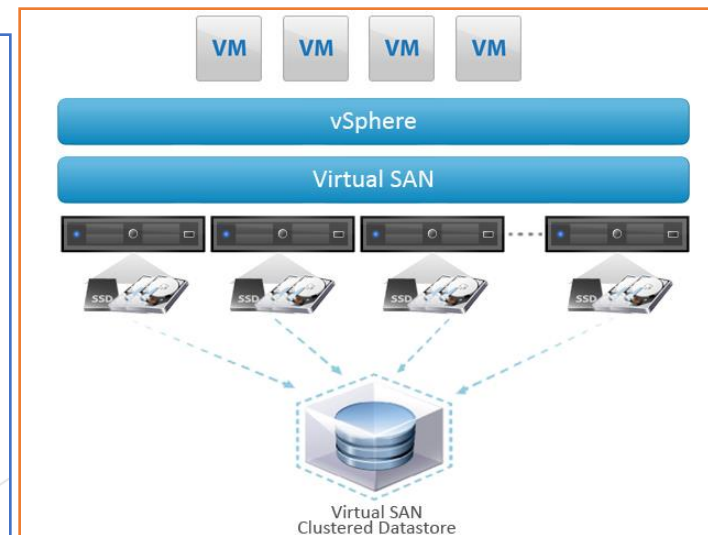
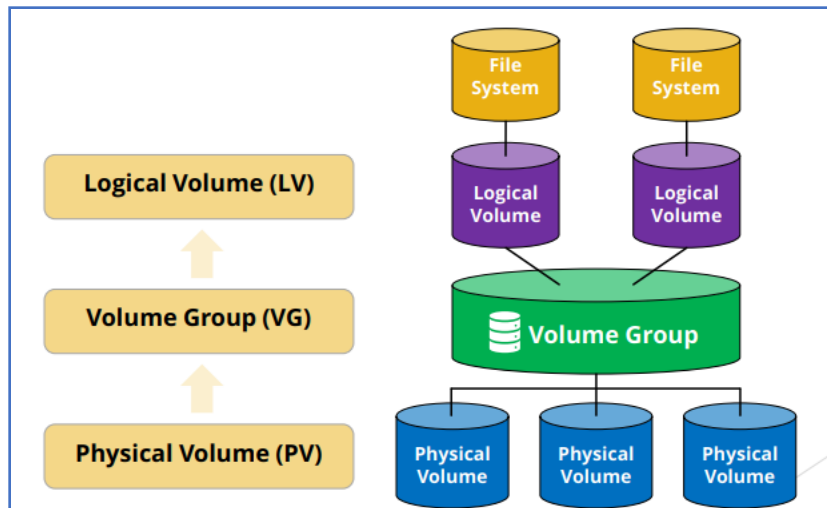
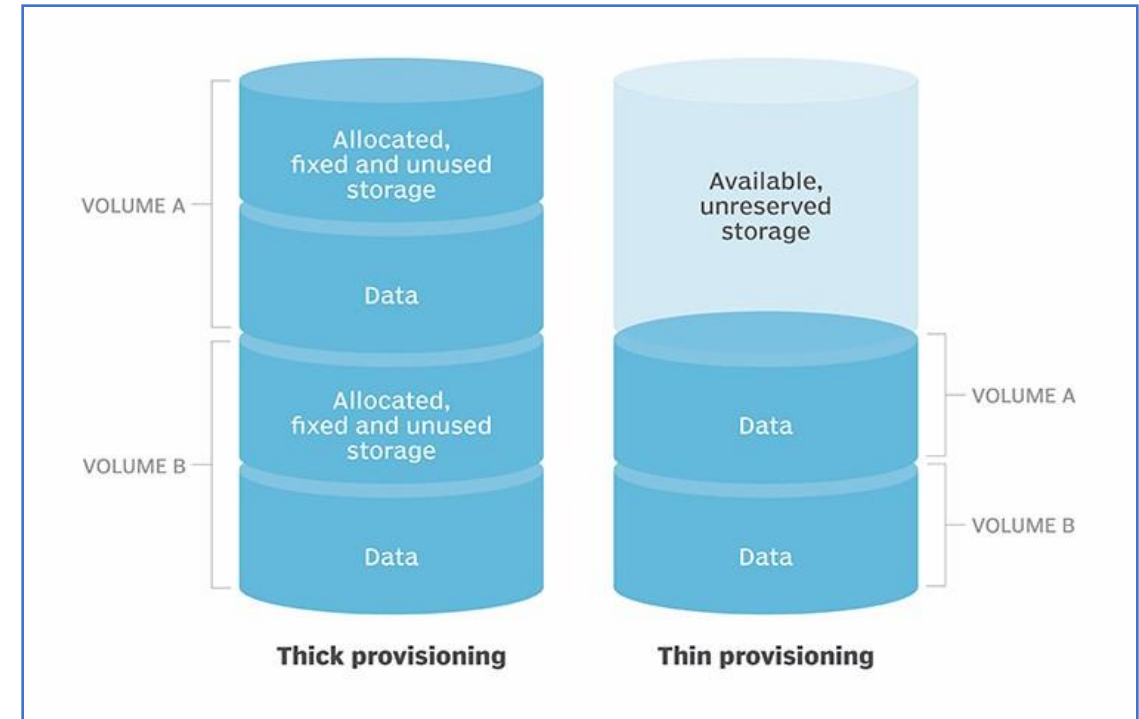
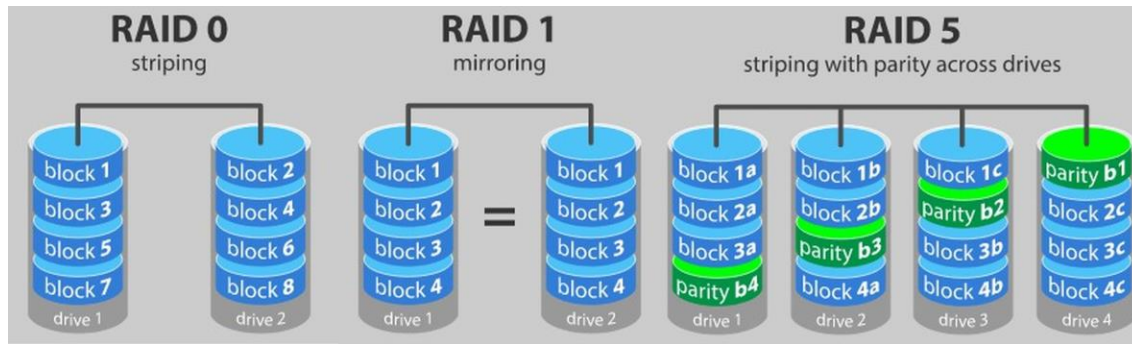


Network Virtualization

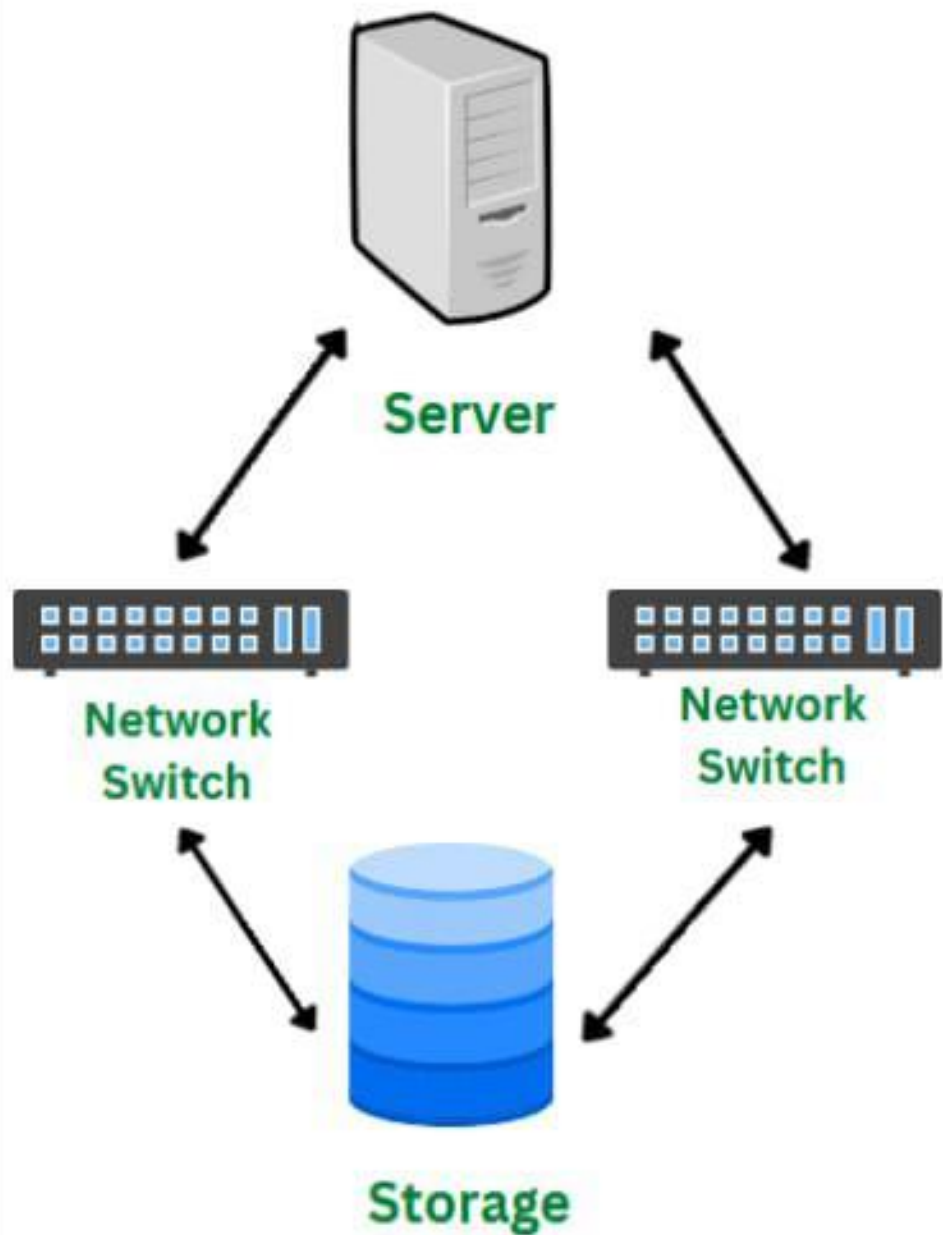
- Logical Partitioning
 - ❖ VLAN, VRF, VDC, Virtual Context etc.
- Network Simplification and Load Balancing
 - ❖ Etherchannel, vPC, Fabricpath
- Management Consolidation and Cabling Optimization
 - ❖ Fabric Extender
- Network Expansion
 - ❖ EoMPLS, VPLS, OTV
- Virtual Networking
 - ❖ vSwitch, VXLAN, VM-FEX
- Virtual Network Services
 - ❖ VSG, Edge Firewall, vWAAS, CSRv, V-Path



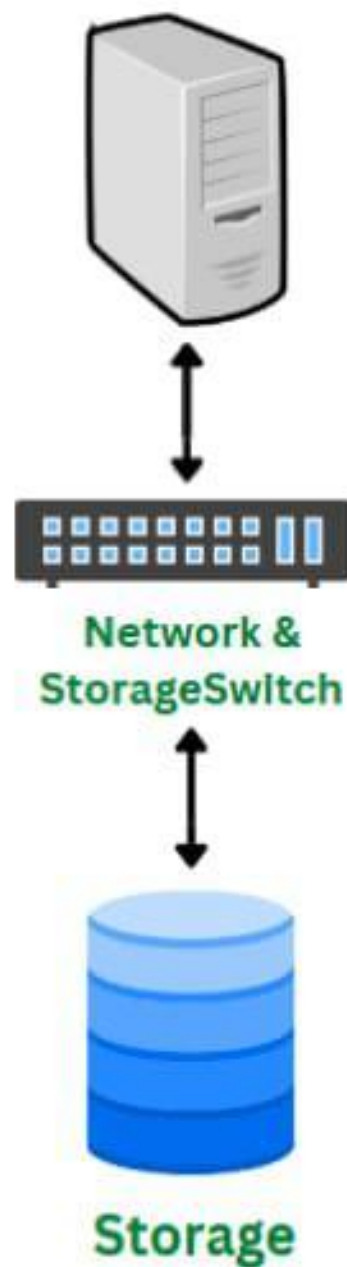
Storage Virtualization



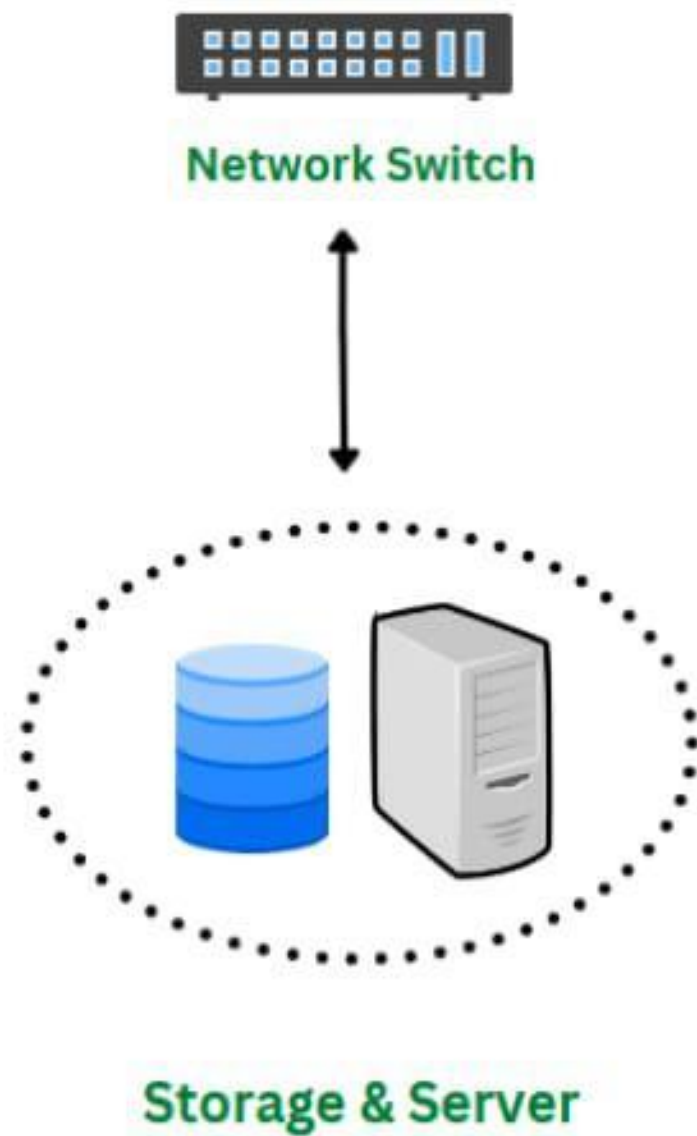
Non-Converged



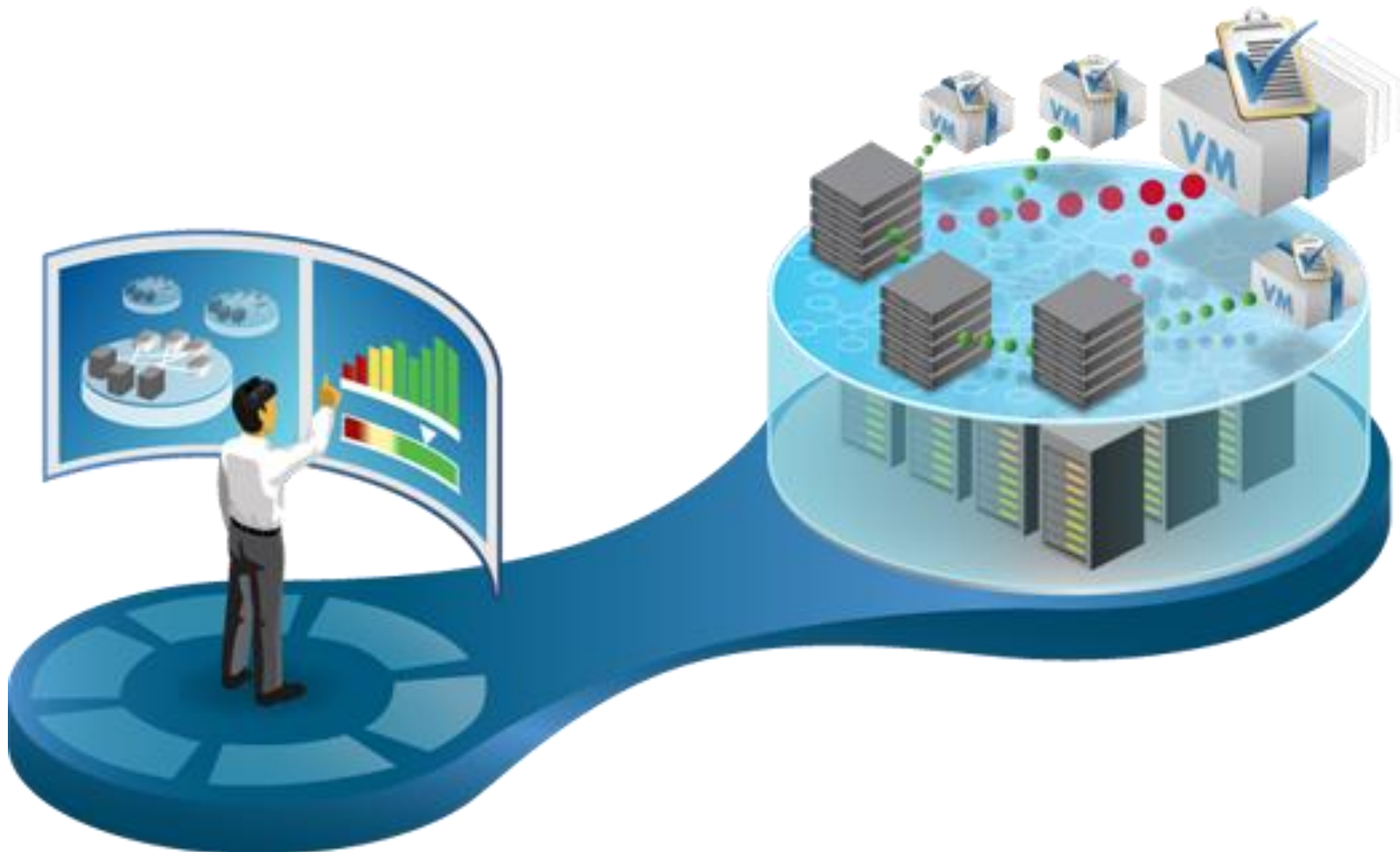
Converged



Hyper-Converged



Virtual Data Center



PROXMOX

INSTALL

&

SETUP



Identity and Access Management



Virtualization Networking and Storage Models



